

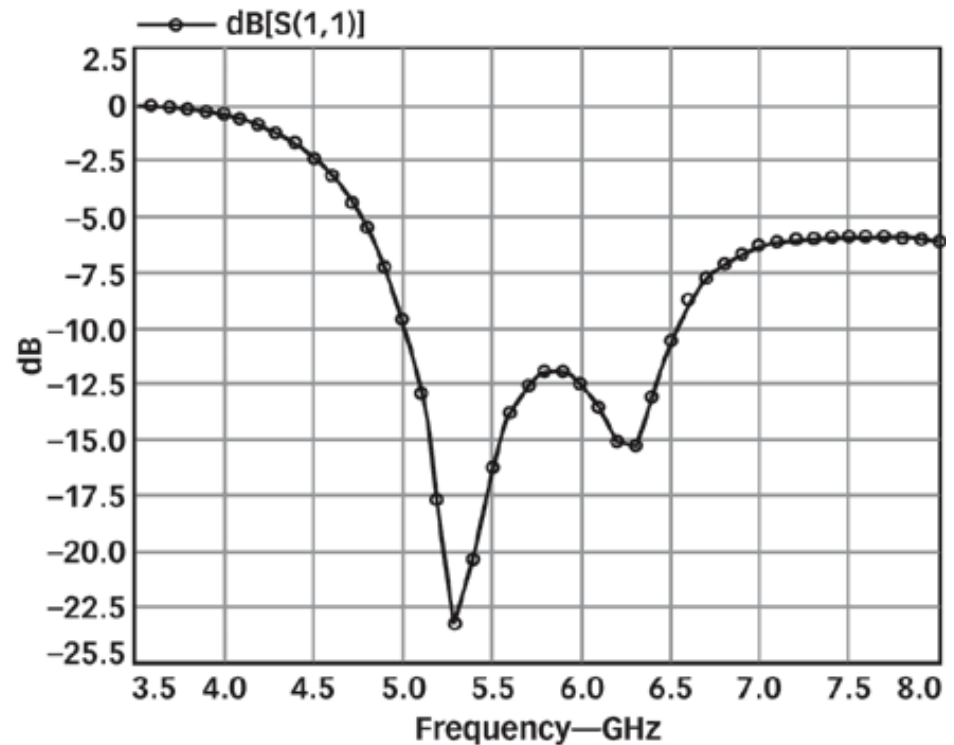
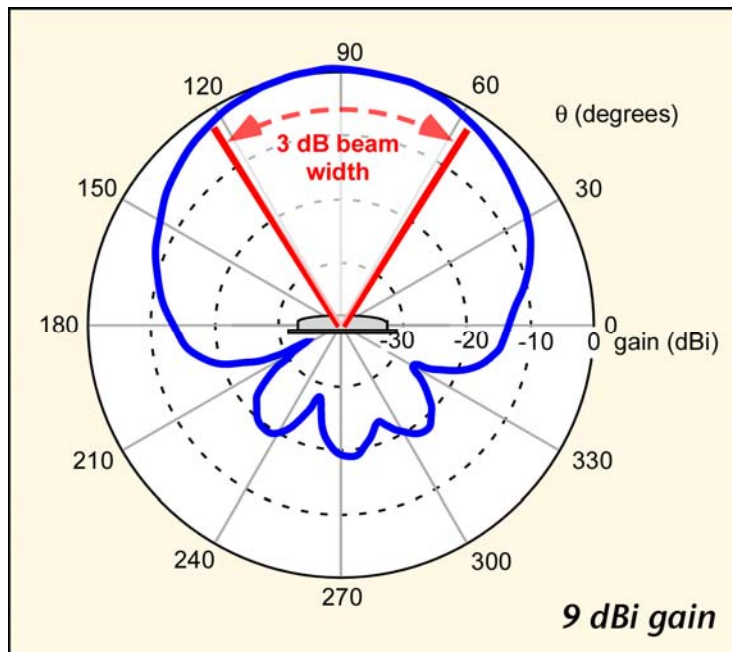
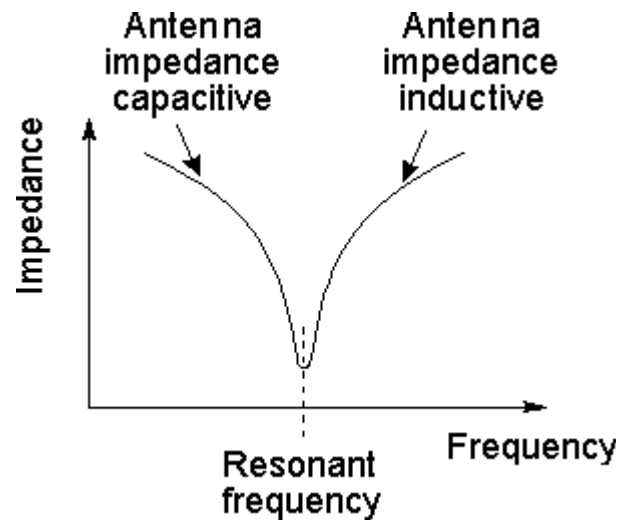
CHƯƠNG 5 MỘT SỐ LOẠI ANTEN

1. Dipole dải rộng
2. Anten Yagi
3. Anten Helic
5. Anten parabol
6. Anten vi dải

1. Dipole dải rộng

❖ Bảng thông của anten

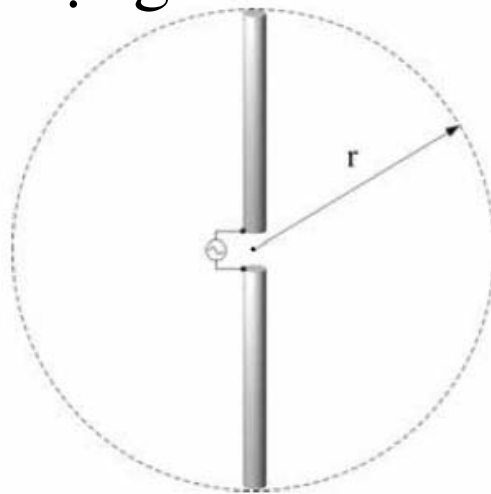
- *Pattern bandwidth*
- *Impedance bandwidth*



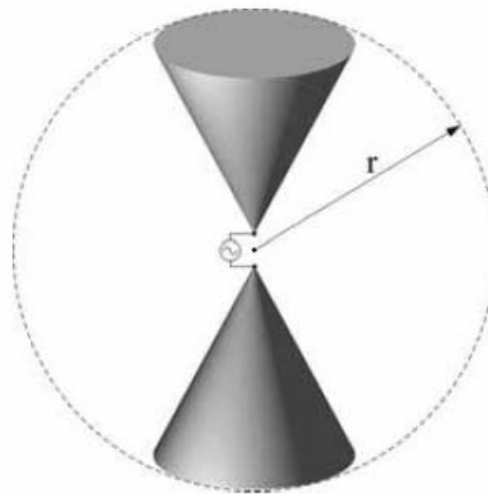
❖ Dipole dải rộng

- Dipole có đường kính lớn
- Dipole dạng nón kép
- Dipole bề vòng

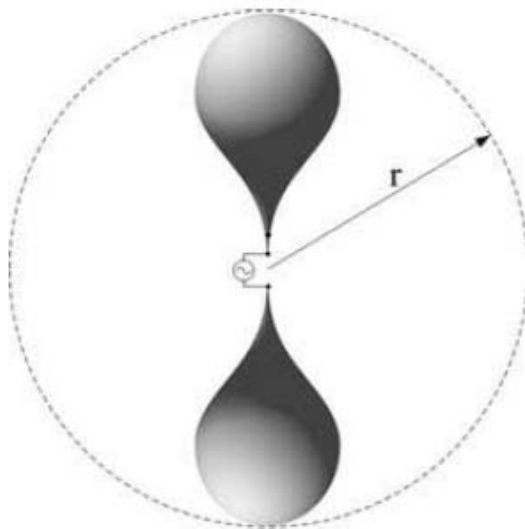
Dipole dải rộng



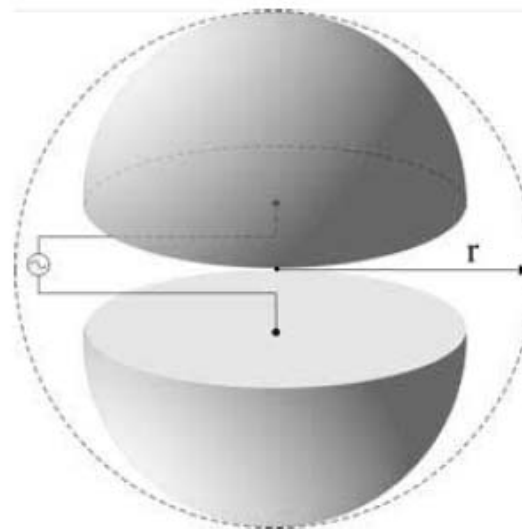
(a) classic (narrow BW)



(b) biconical (intermediate BW)

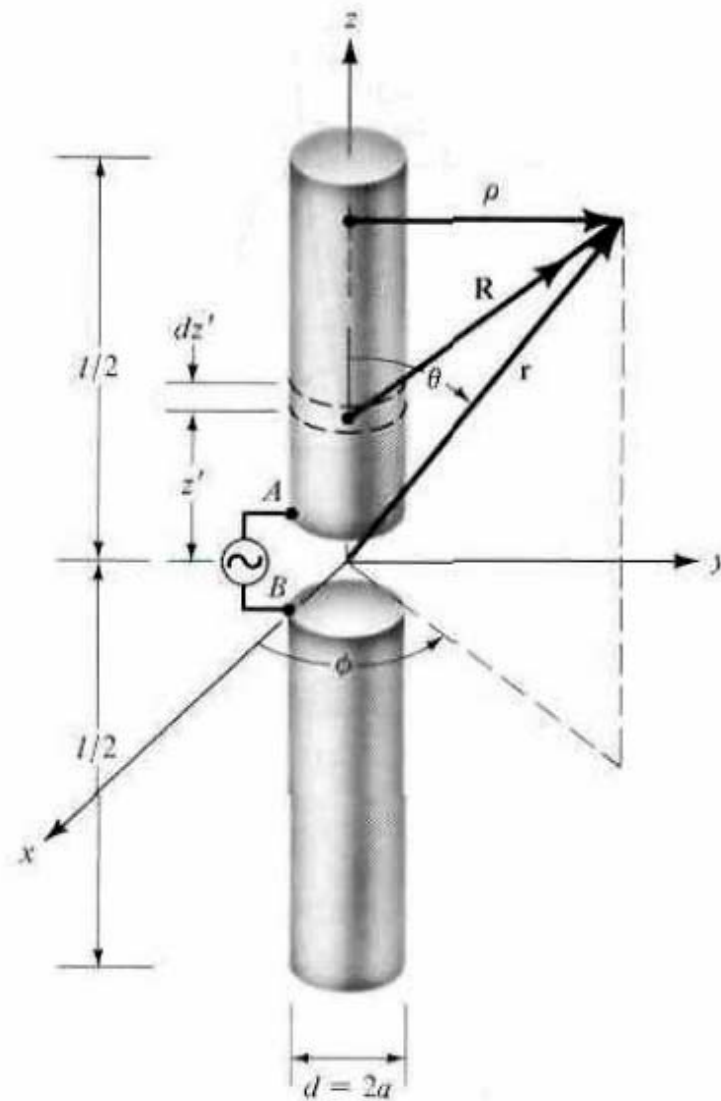


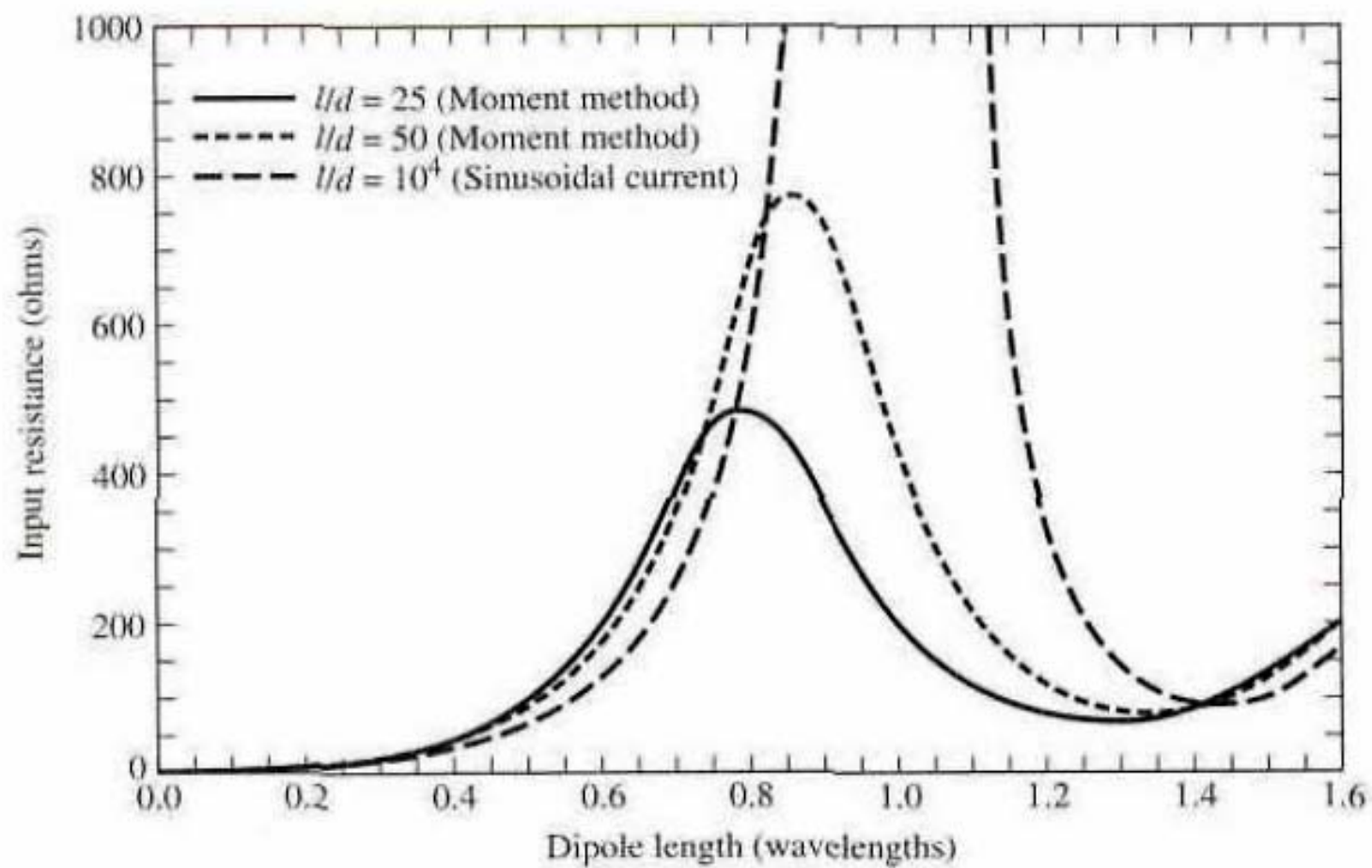
(c) tapered (intermediate BW)



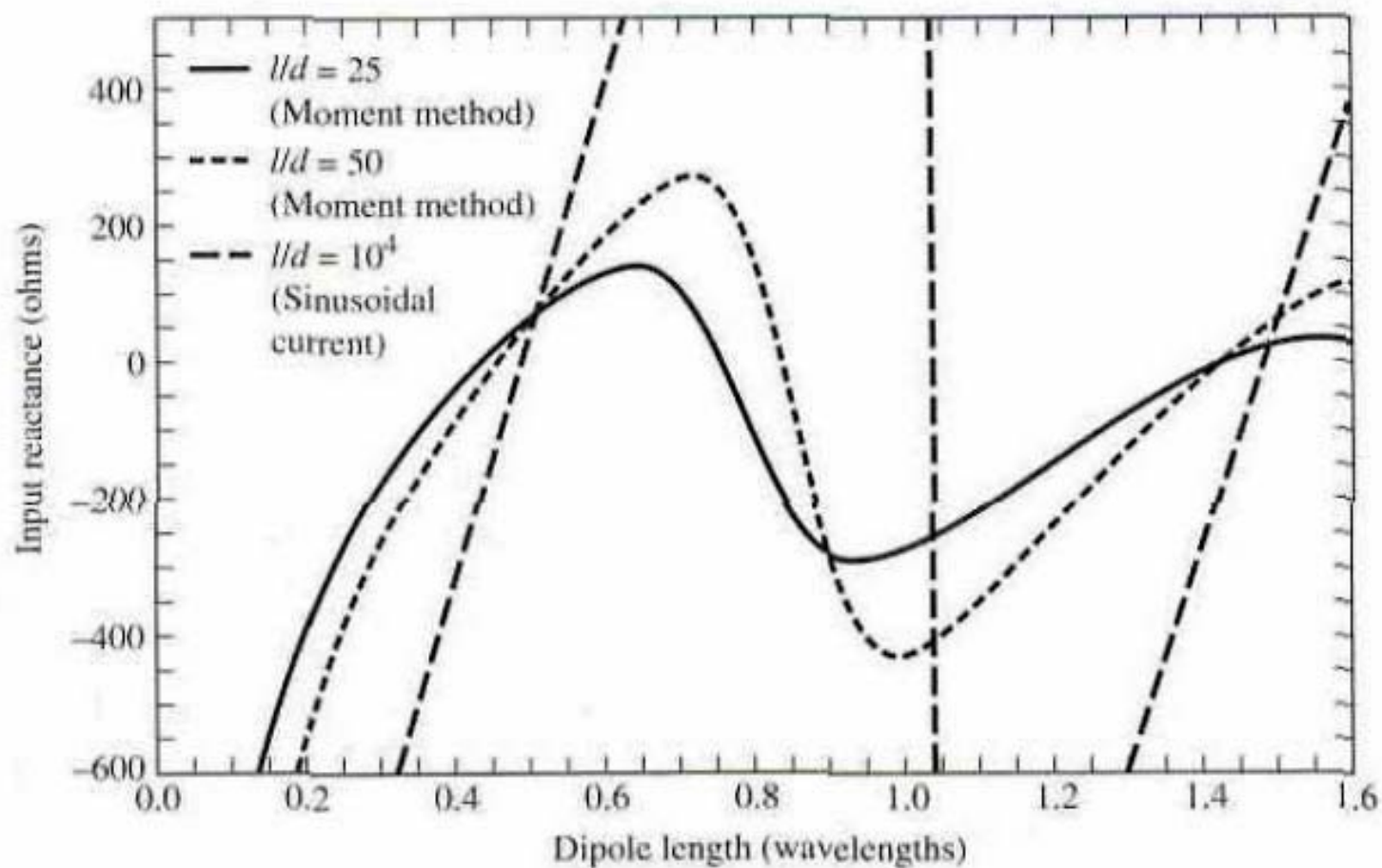
(d) hemispherical (wide BW)

➤ Dipole có đường kính lớn





(a) Input resistance



(b) Input reactance

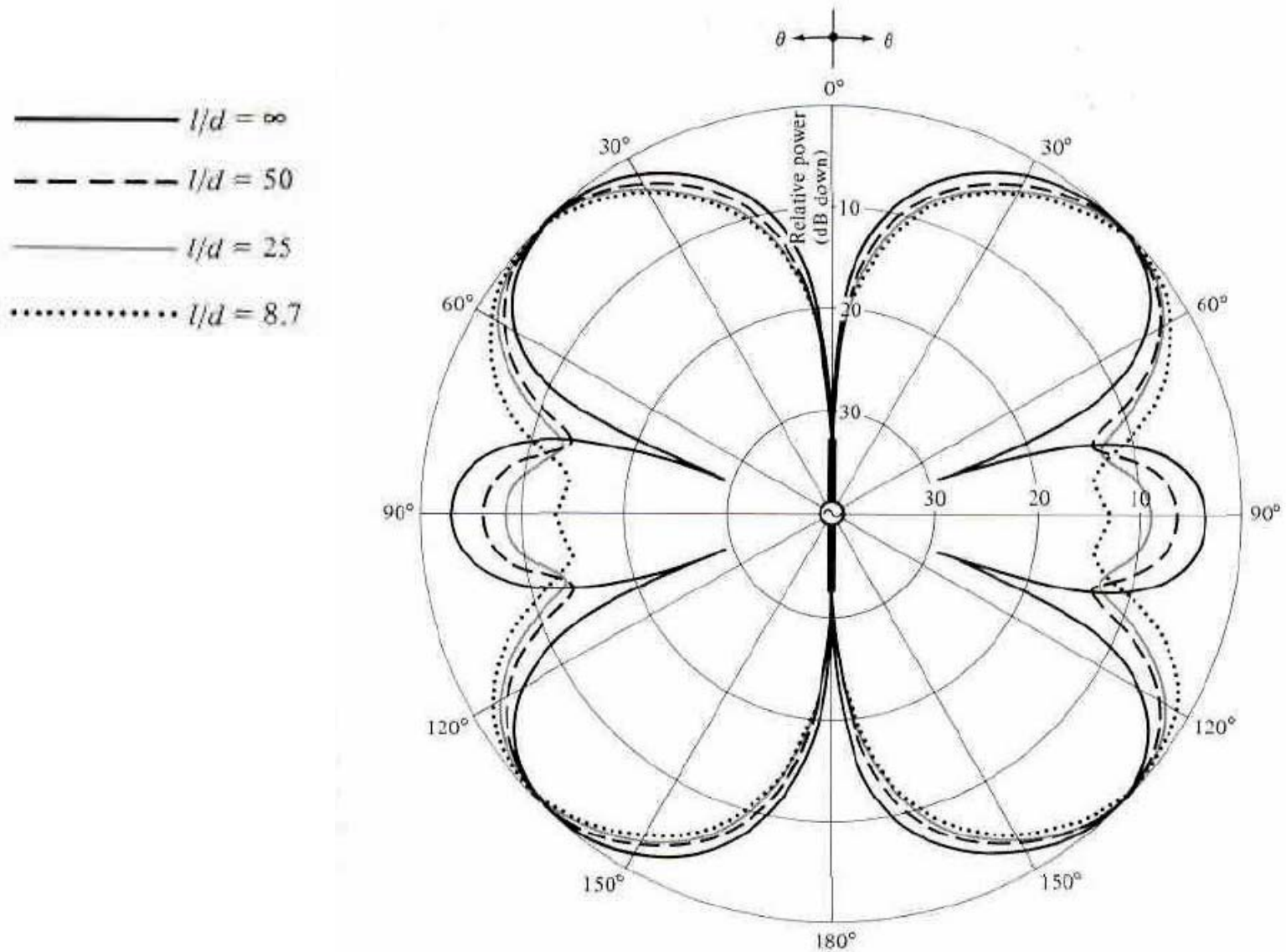
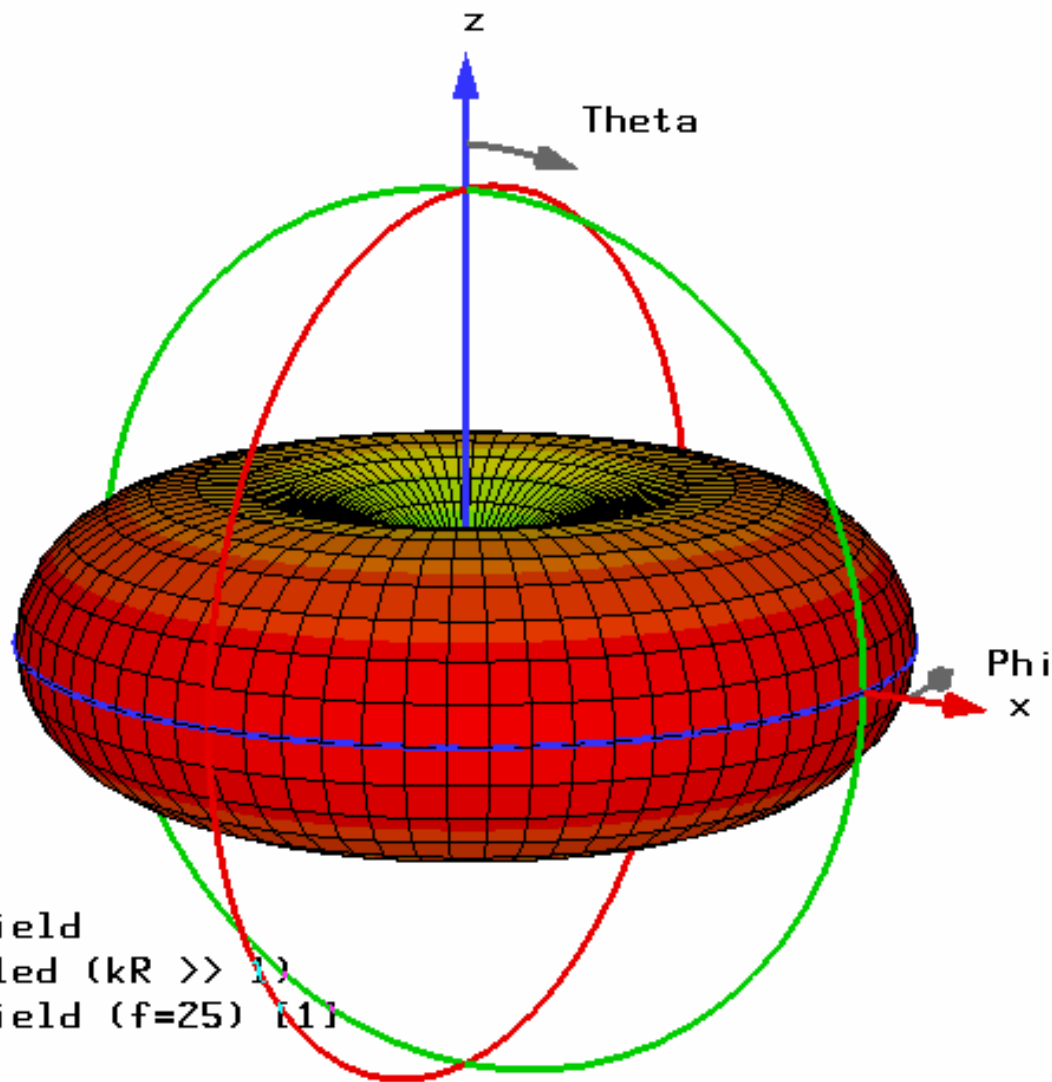


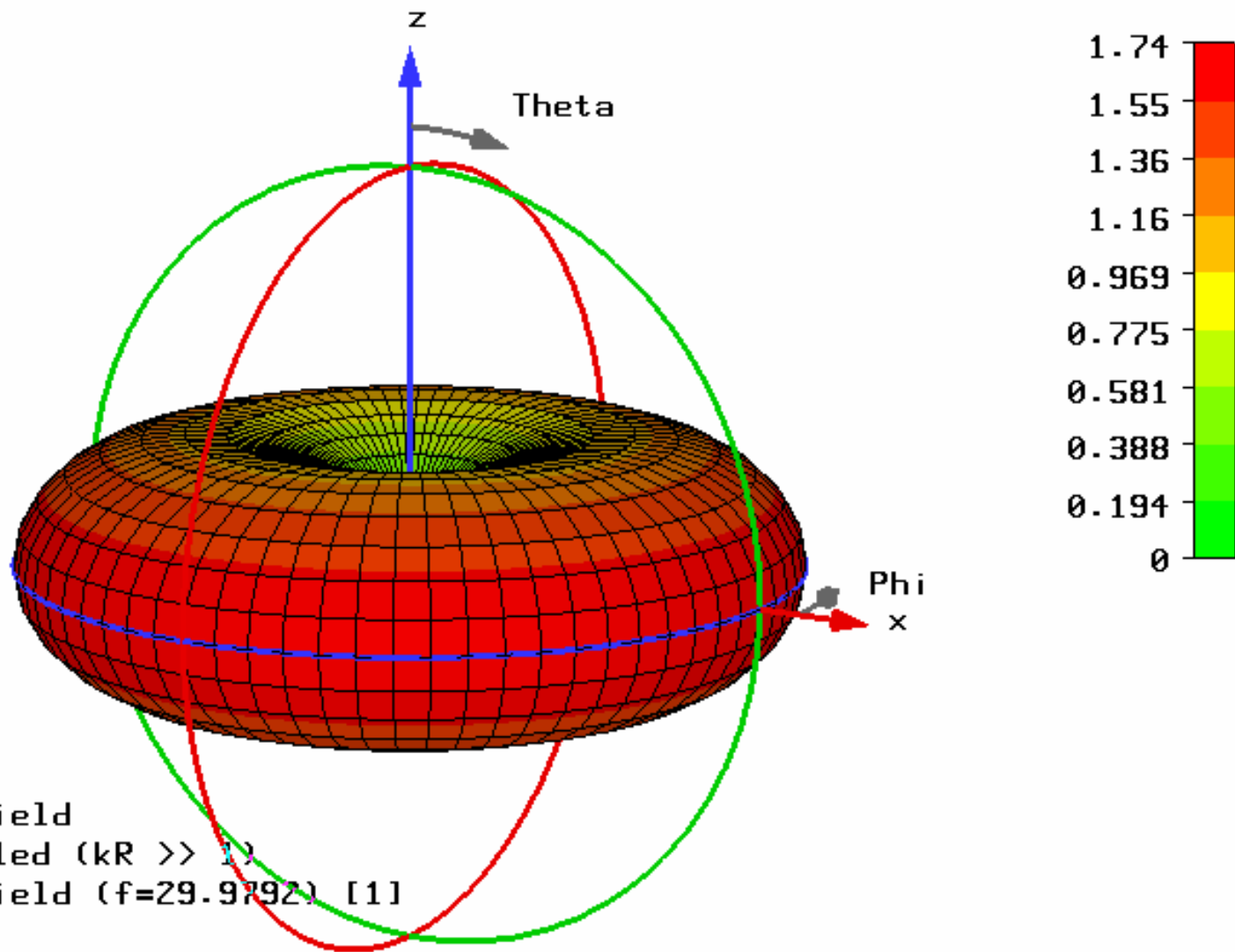
Figure 9.10 Amplitude radiation patterns of a $3\lambda/2$ dipole of various thicknesses.

$$l < \frac{\lambda}{2}$$



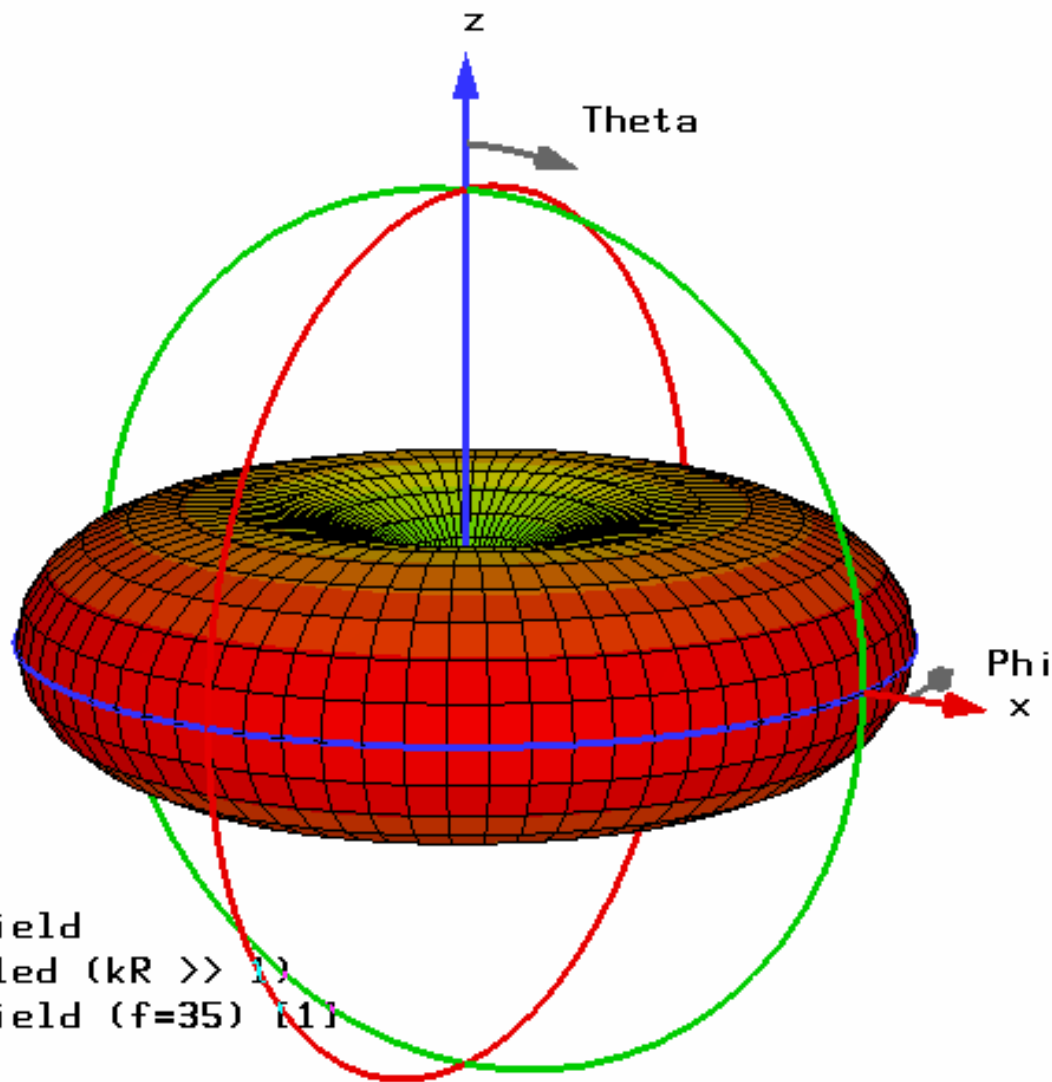
Type = Farfield
 Approximation = enabled ($kR \gg 1$)
 Monitor = farfield (f=25) [1]
 Component = Abs
 Output = Directivity
 Frequency = 25
 Rad. effic. = 0.9455
 Tot. effic. = 0.5697
 Dir. = 1.658

$$l = \frac{\lambda}{2}$$



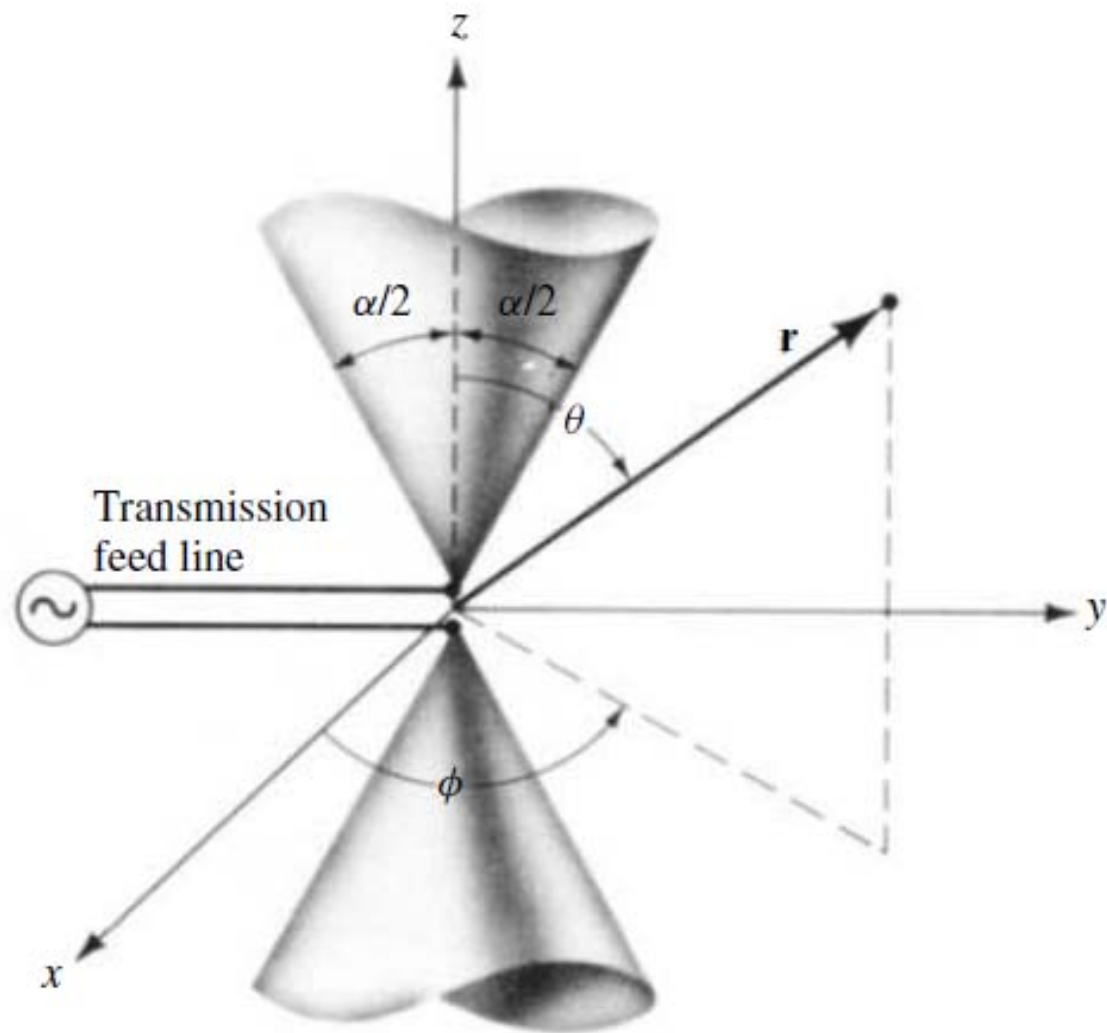
Type = Farfield
 Approximation = enabled ($kR \gg 1$)
 Monitor = farfield (f=29.9792) [1]
 Component = Abs
 Output = Directivity
 Frequency = 29.9792
 Rad. effic. = 1.109
 Tot. effic. = 0.8787
 Dir. = 1.744

$$l > \frac{\lambda}{2}$$



Type = Farfield
 Approximation = enabled ($kR \gg 1$)
 Monitor = farfield (f=35) [1]
 Component = Abs
 Output = Directivity
 Frequency = 35
 Rad. effic. = 1.020
 Tot. effic. = 0.7696
 Dir. = 1.782

➤ Dipole dạng nón kép



(a) Biconical geometry

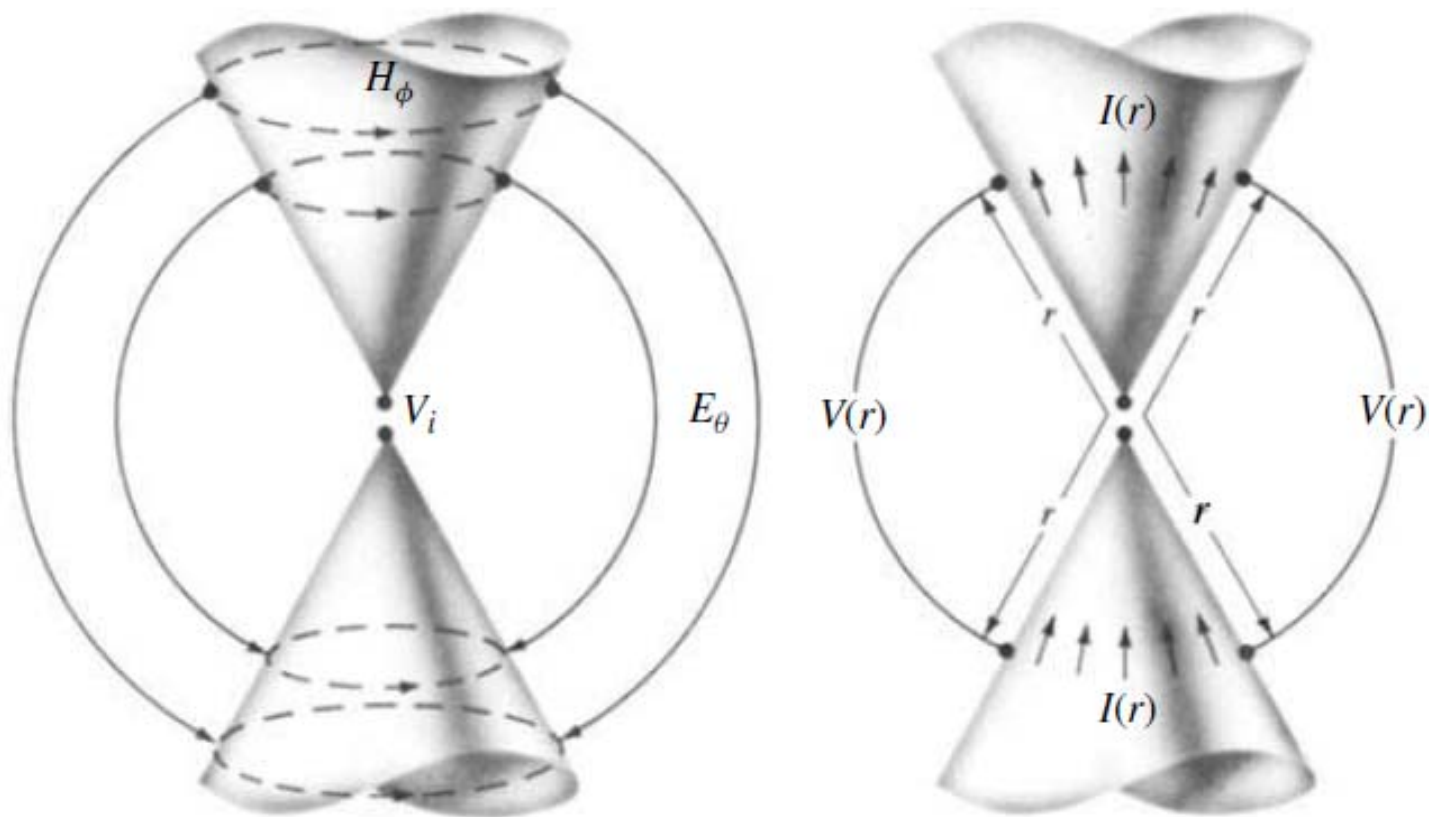
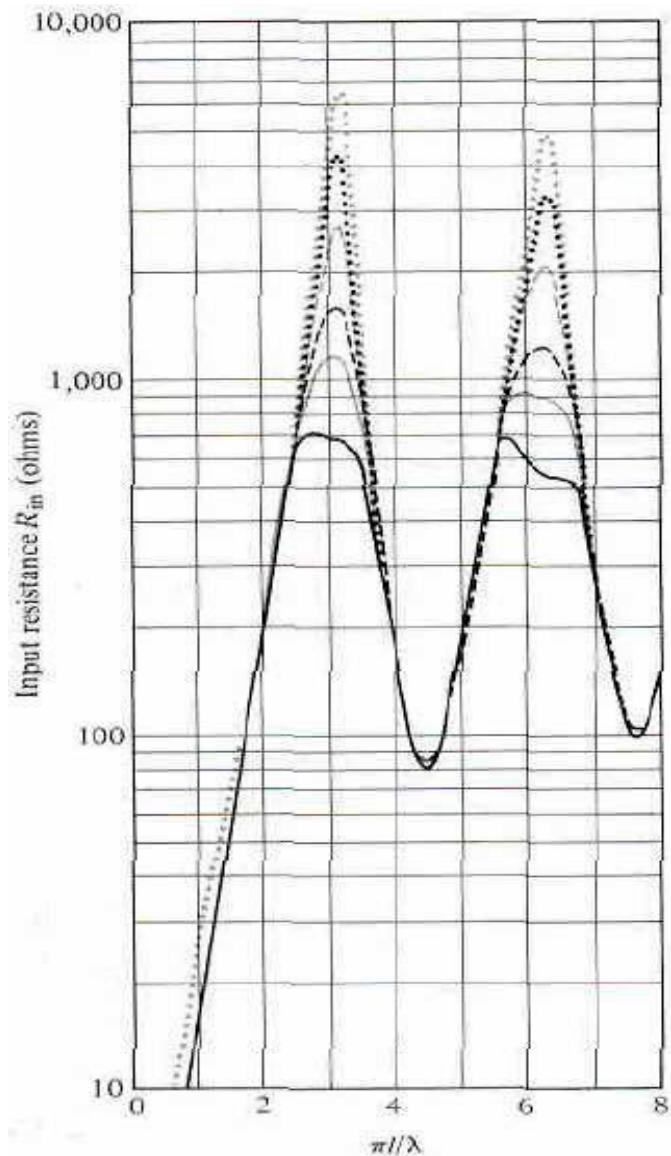


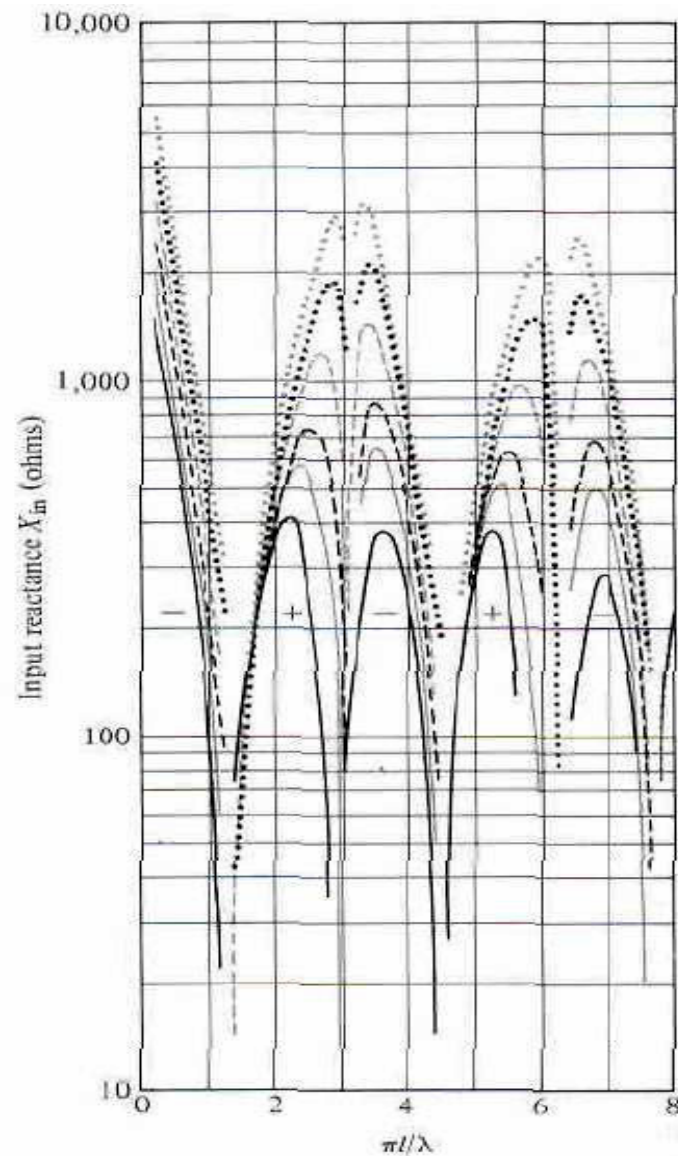
Figure 9.4 Electric and magnetic fields, and associated voltages and currents, for a biconical antenna.

Trở kháng vào của
dipole nón kép có
chiều dài hữu hạn



— $\alpha/2 = 5^\circ$
 - - - $\alpha/2 = 2^\circ$
 - · - · $\alpha/2 = 1^\circ$

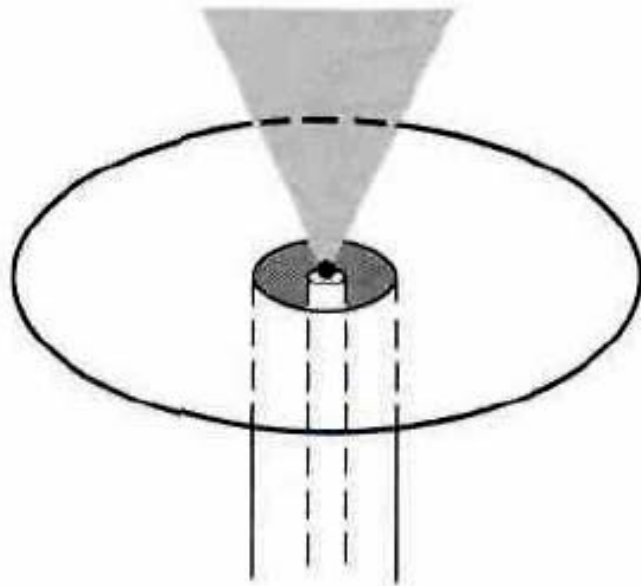
(a) Resistance



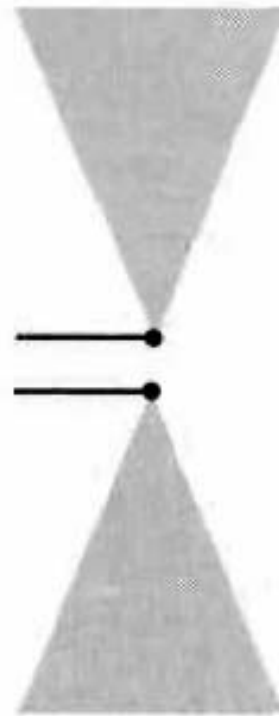
- - - $\alpha/2 = \frac{1}{4}^\circ$
 ····· $\alpha/2 = \frac{1}{20}^\circ$
 ····· $\alpha/2 = \frac{1}{100}^\circ$

(b) Reactance

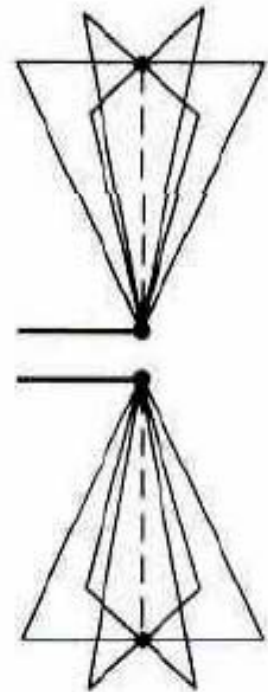
Một số dạng dipole nón kép cải biên



(a) Triangular sheet



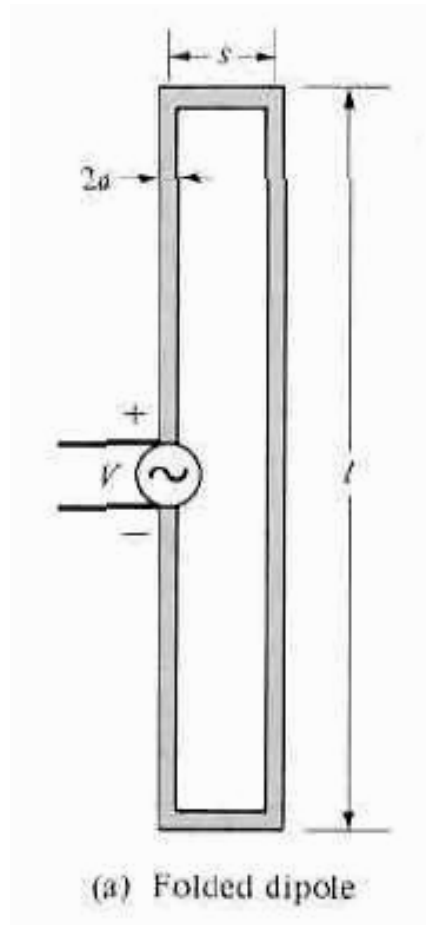
(b) Bow-tie



(c) Wire simulation

Figure 9.5 Triangular sheet, bow-tie, and wire simulation of biconical antenna.

➤ Dipole bề vòng



Trở kháng vào của dipole bề vòng

Trường hợp: $l = \frac{\lambda}{2}$

Mode đường truyền sóng:

$$Z_t \rightarrow \infty$$

$$\Rightarrow I_t = 0$$

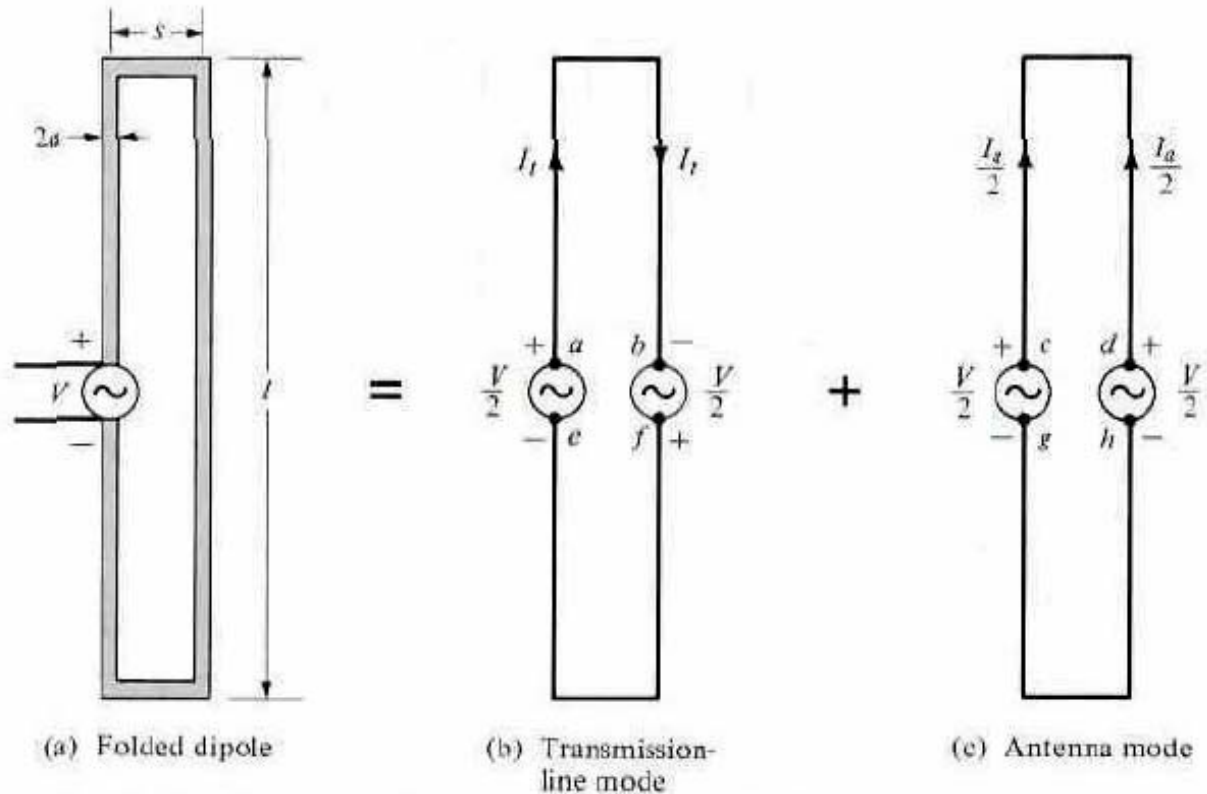
Mode anten:

$$I_a = \frac{V/2}{Z_{dipole}}$$

I tổng:

$$I_{in} = I_t + \frac{I_a}{2} = 0 + \frac{V/2}{2.Z_{dipole}}$$

$$\Rightarrow Z_{in} = \frac{V}{I_{in}} = 4.Z_{dipole}$$



Trường hợp tổng quát:

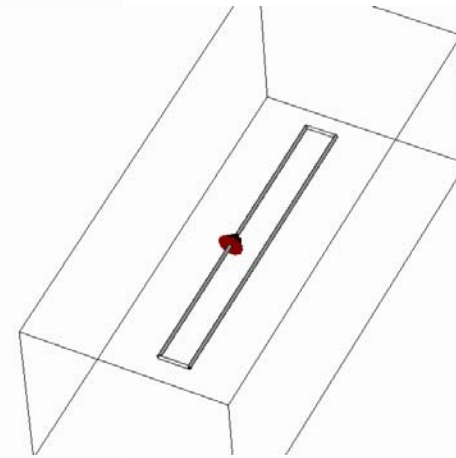
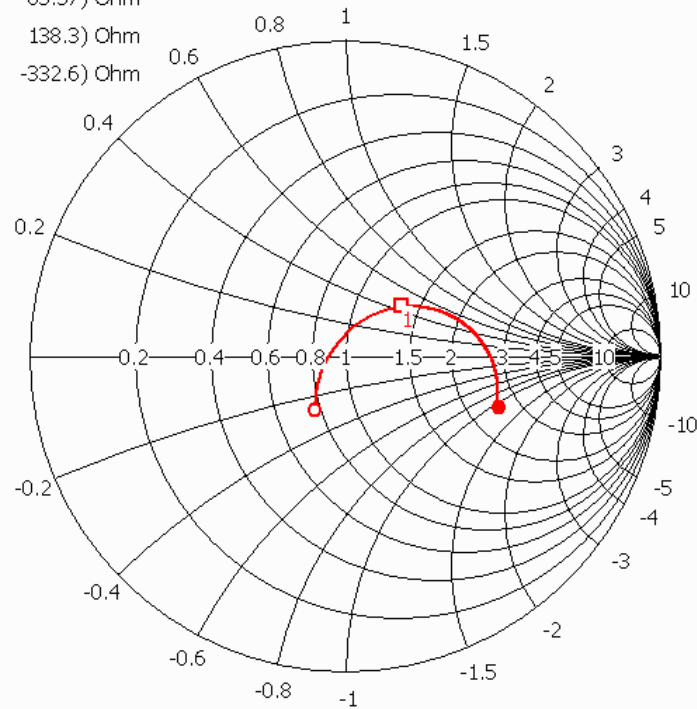
$$Z_{in} = \frac{V}{I_{in}} = \frac{2Z_t(4Z_d)}{2Z_t + 4Z_d} = \frac{4Z_t Z_d}{2Z_d + Z_t}$$

○
□
●

25.00 (233.1, -83.57) Ohm
28.73 (406.4, 138.3) Ohm
35.00 (760.4, -332.6) Ohm

S-Parameter Smith Chart

S1,1 (300 Ohm)

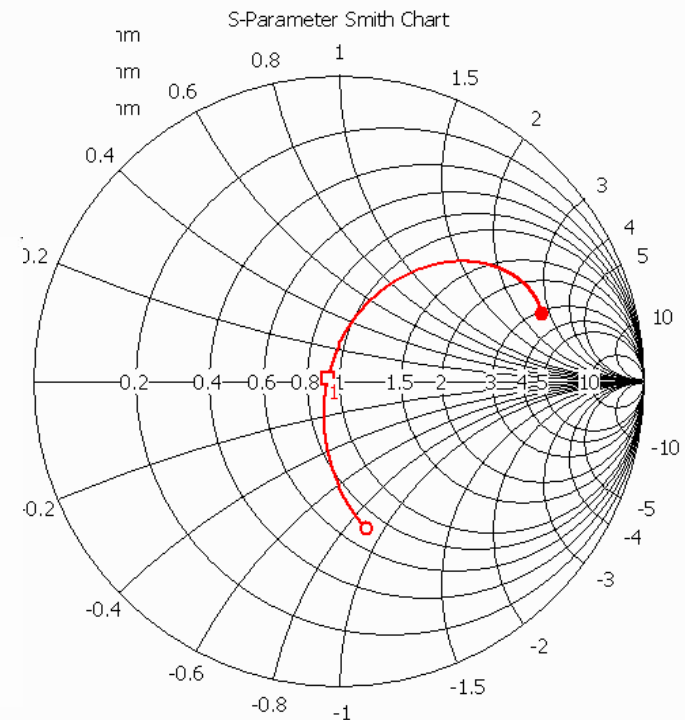
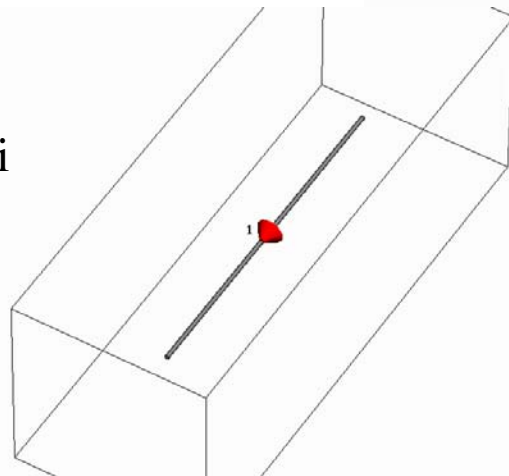


Dipole bề
vòng được
nối với nguồn
có trở kháng

$$Z_S = 300\Omega$$

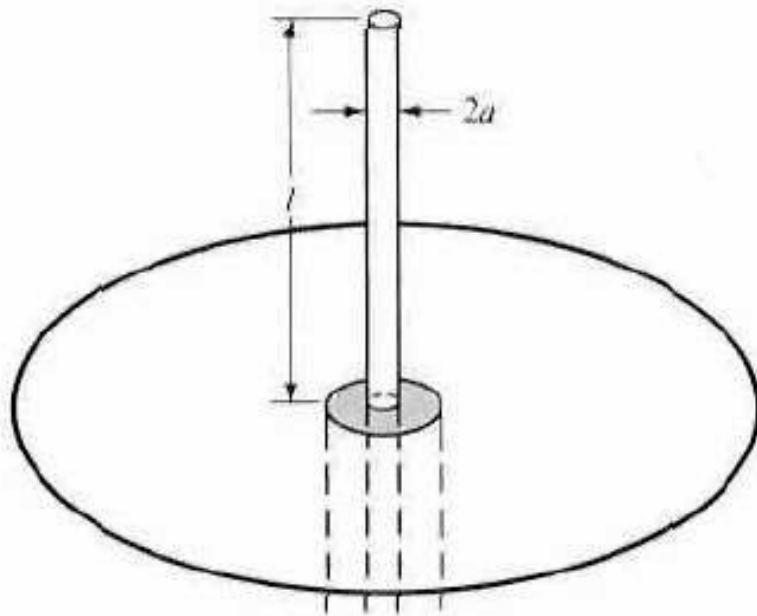
Dipole được nối với
nguồn có trở kháng

$$Z_S = 75\Omega$$

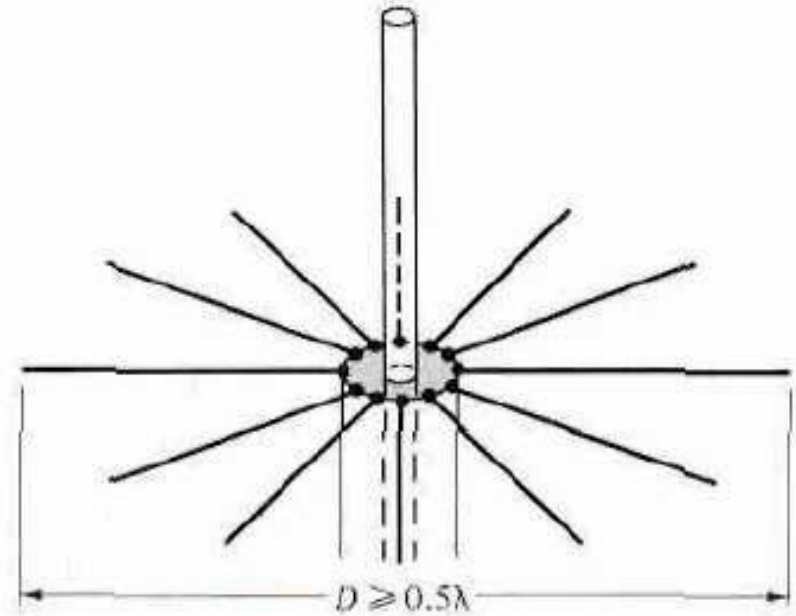


S1,1 (75 Ohm)

Một số dạng Monopole

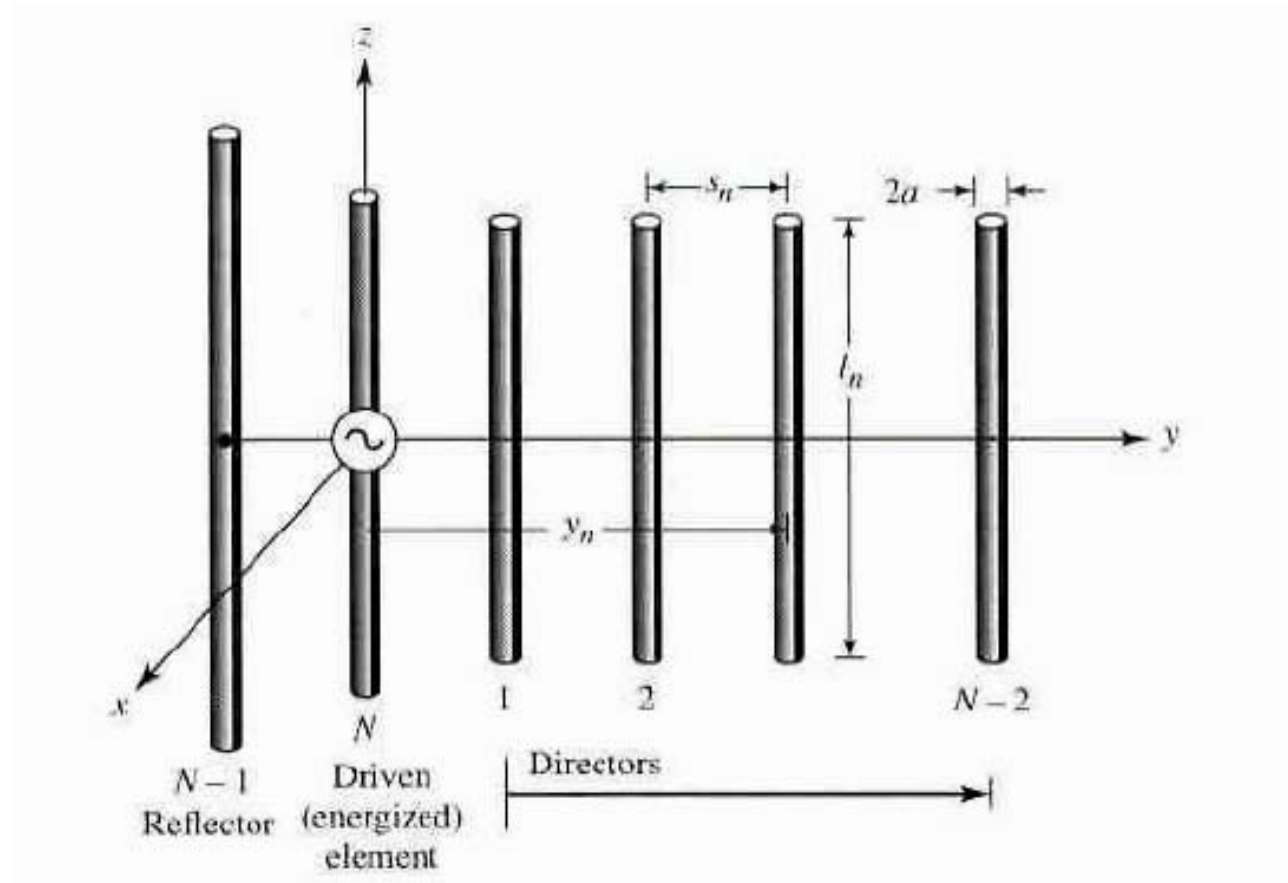


(a) Cylindrical monopole



(b) Wire simulation of ground plane

2. Anten Yagi



Thông thường phần tử chủ động cộng hưởng tương ứng với chiều dài $0,45 \div 0,49\lambda$
Có dạng dipole thường hoặc dipole bề vòng.

Trong khi đó các phần tử hướng xạ có chiều dài khoảng $0,4 \div 0,45\lambda$
và chúng không nhất thiết phải có chiều dài bằng nhau.

Khoảng cách giữa các phần tử hướng xạ khoảng $0,3 \div 0,4\lambda$ và chúng
cũng không nhất thiết phải cách đều nhau.

Chiều dài phần tử phản xạ lớn hơn phần tử chủ động và nó cách phần
tử chủ động khoảng $0,25\lambda$

**Table 10.6 OPTIMIZED UNCOMPENSATED LENGTHS OF PARASITIC ELEMENTS
FOR YAGI-UDA ANTENNAS OF SIX DIFFERENT LENGTHS**

$d/\lambda = 0.0085$ $s_{12} = 0.2\lambda$		LENGTH OF YAGI-UDA (IN WAVELENGTHS)					
		0.4	0.8	1.20	2.2	3.2	4.2
LENGTH OF REFLECTOR (l_1/λ)		0.482	0.482	0.482	0.482	0.482	0.475
LENGTH OF DIRECTORS, λ	l_3	0.442	0.428	0.428	0.432	0.428	0.424
	l_4		0.424	0.420	0.415	0.420	0.424
	l_5		0.428	0.420	0.407	0.407	0.420
	l_6			0.428	0.398	0.398	0.407
	l_7				0.390	0.394	0.403
	l_8				0.390	0.390	0.398
	l_9				0.390	0.386	0.394
	l_{10}				0.390	0.386	0.390
	l_{11}				0.398	0.386	0.390
	l_{12}				0.407	0.386	0.390
	l_{13}					0.386	0.390
	l_{14}					0.386	0.390
	l_{15}					0.386	0.390
	l_{16}					0.386	
	l_{17}					0.386	
SPACING BETWEEN DIRECTORS (s_{ik}/λ)		0.20	0.20	0.25	0.20	0.20	0.308
DIRECTIVITY RELATIVE TO HALF-WAVE DIPOLE (dB)		7.1	9.2	10.2	12.25	13.4	14.2
DESIGN CURVE (SEE FIGURE 10.25)		(A)	(B)	(B)	(C)	(B)	(D)

Xét một anten Yagi:

N = total number of elements = 15

number of directors = 13

number of reflectors = 1

number of exciters = 1

total length of reflector = 0.5λ

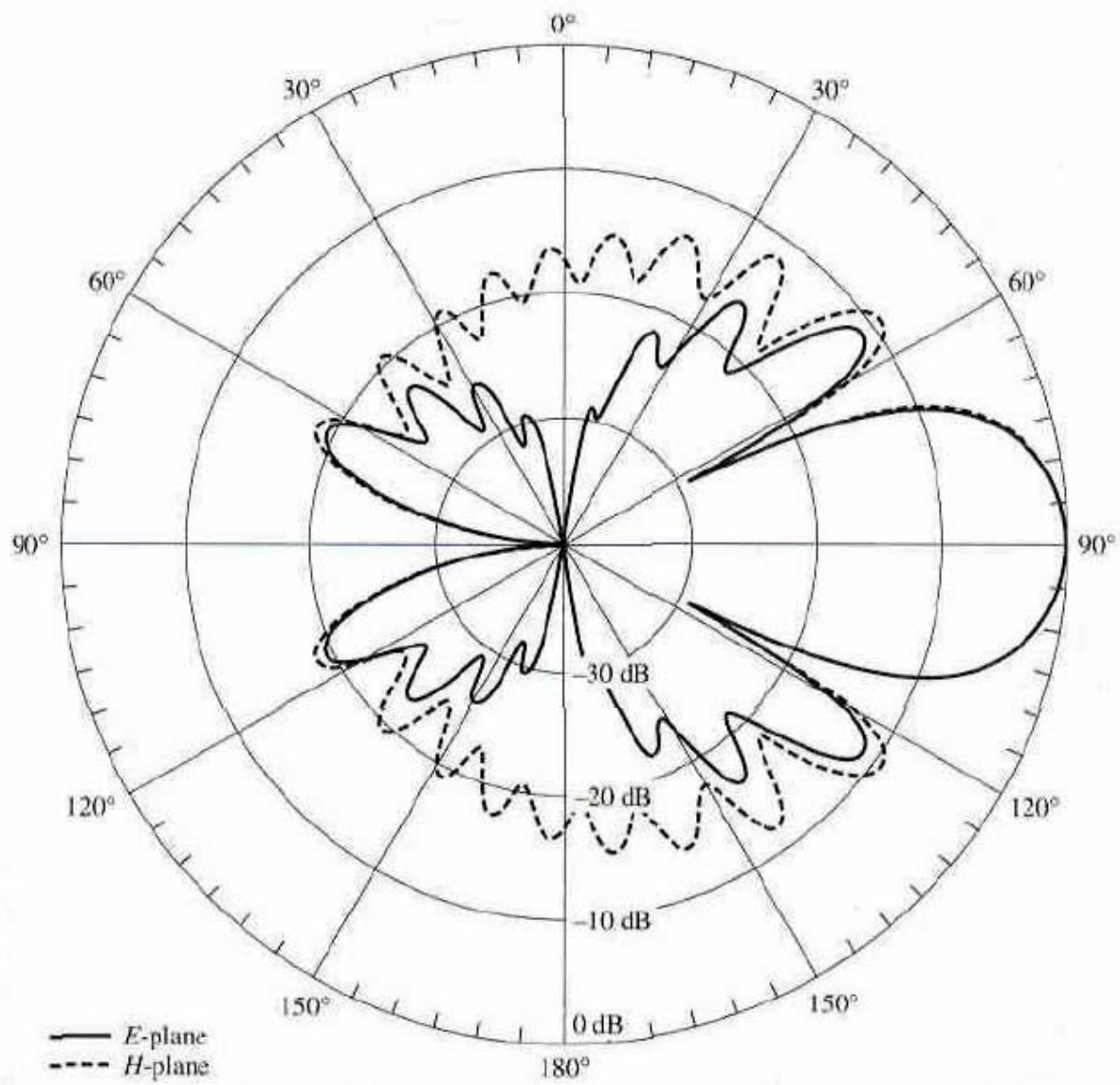
total length of feeder = 0.47λ

total length of each director = 0.406λ

spacing between reflector and feeder = 0.25λ

spacing between adjacent directors = 0.34λ

a = radius of wires = 0.003λ



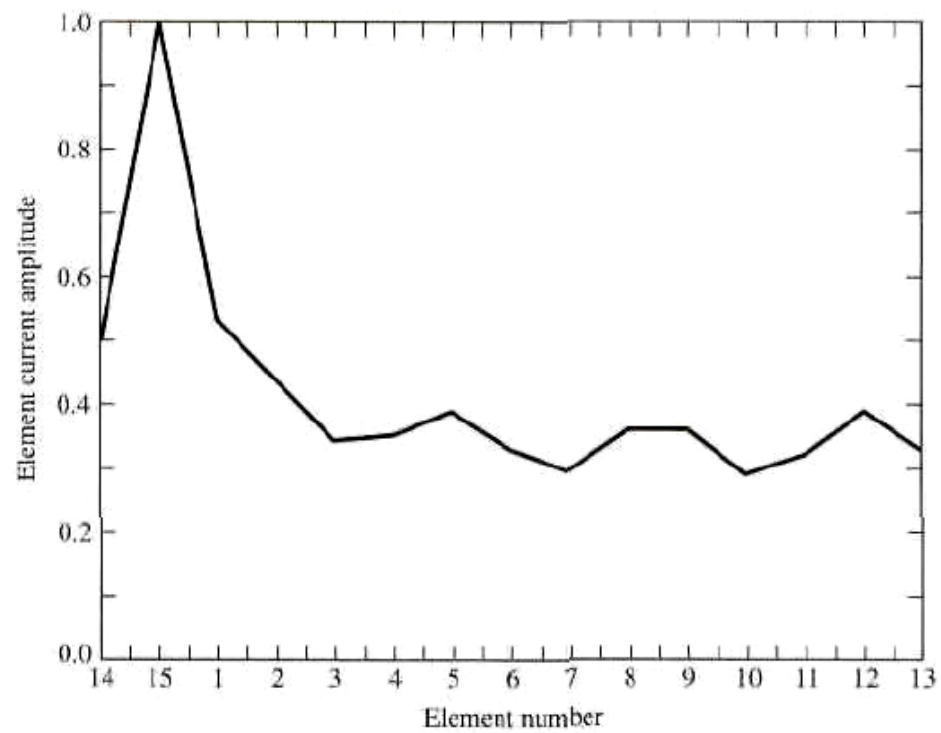


Figure 10.20 Normalized current at the center of each element of a 15-element Yagi-Uda array.

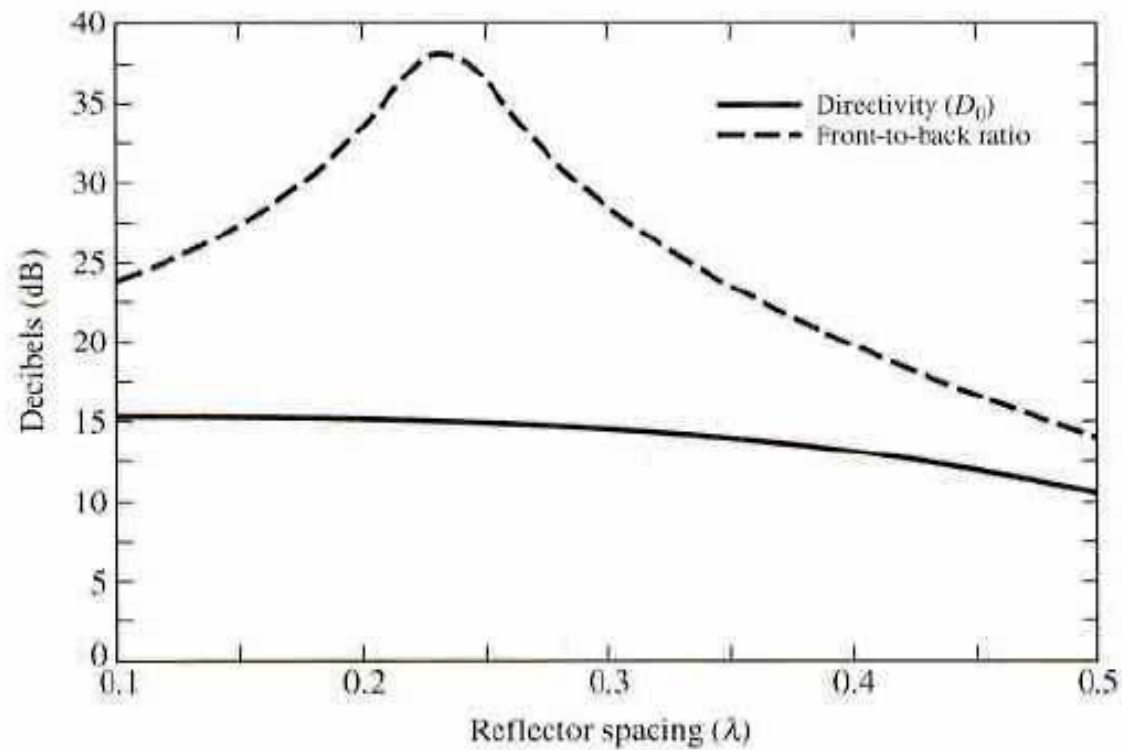
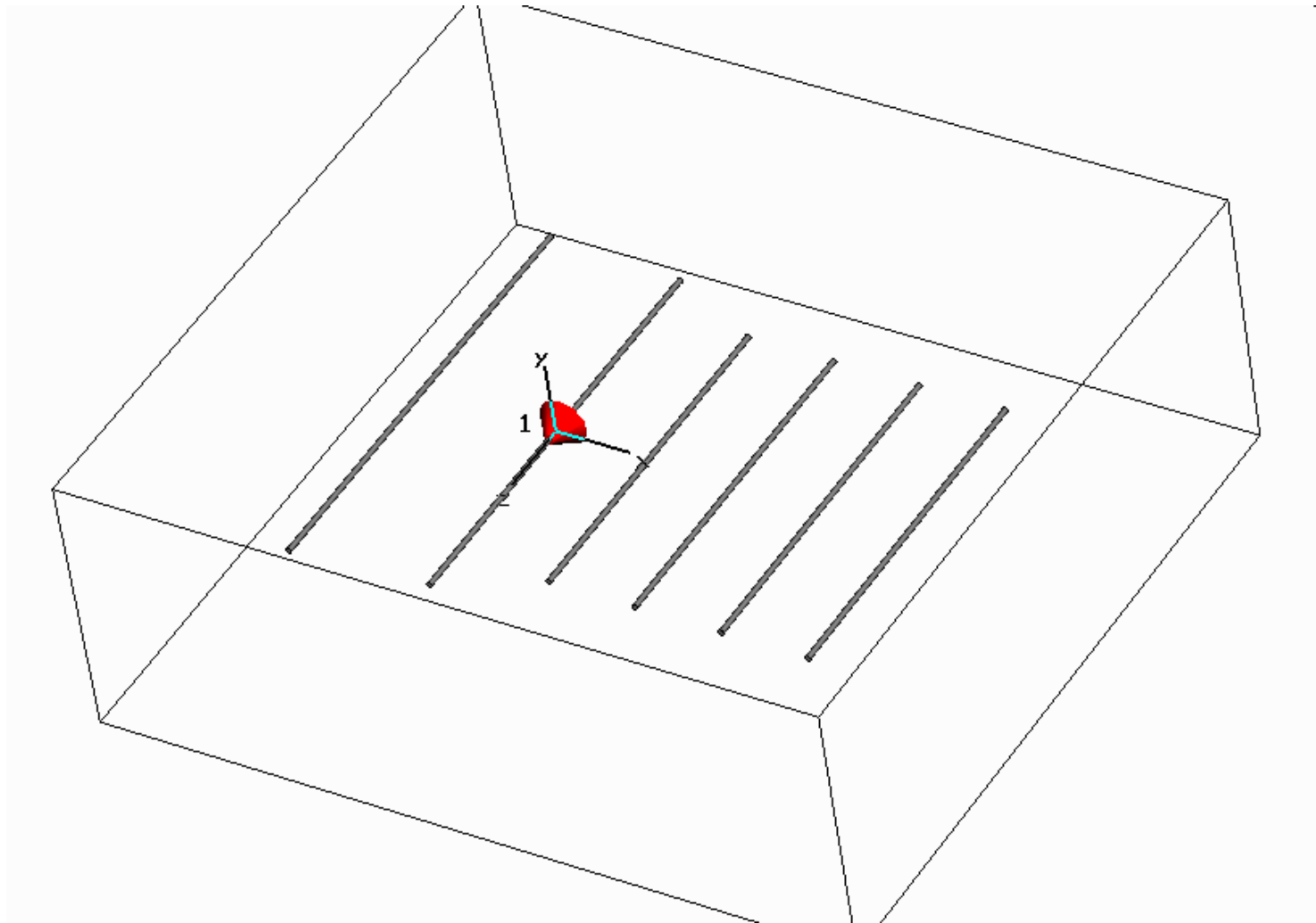
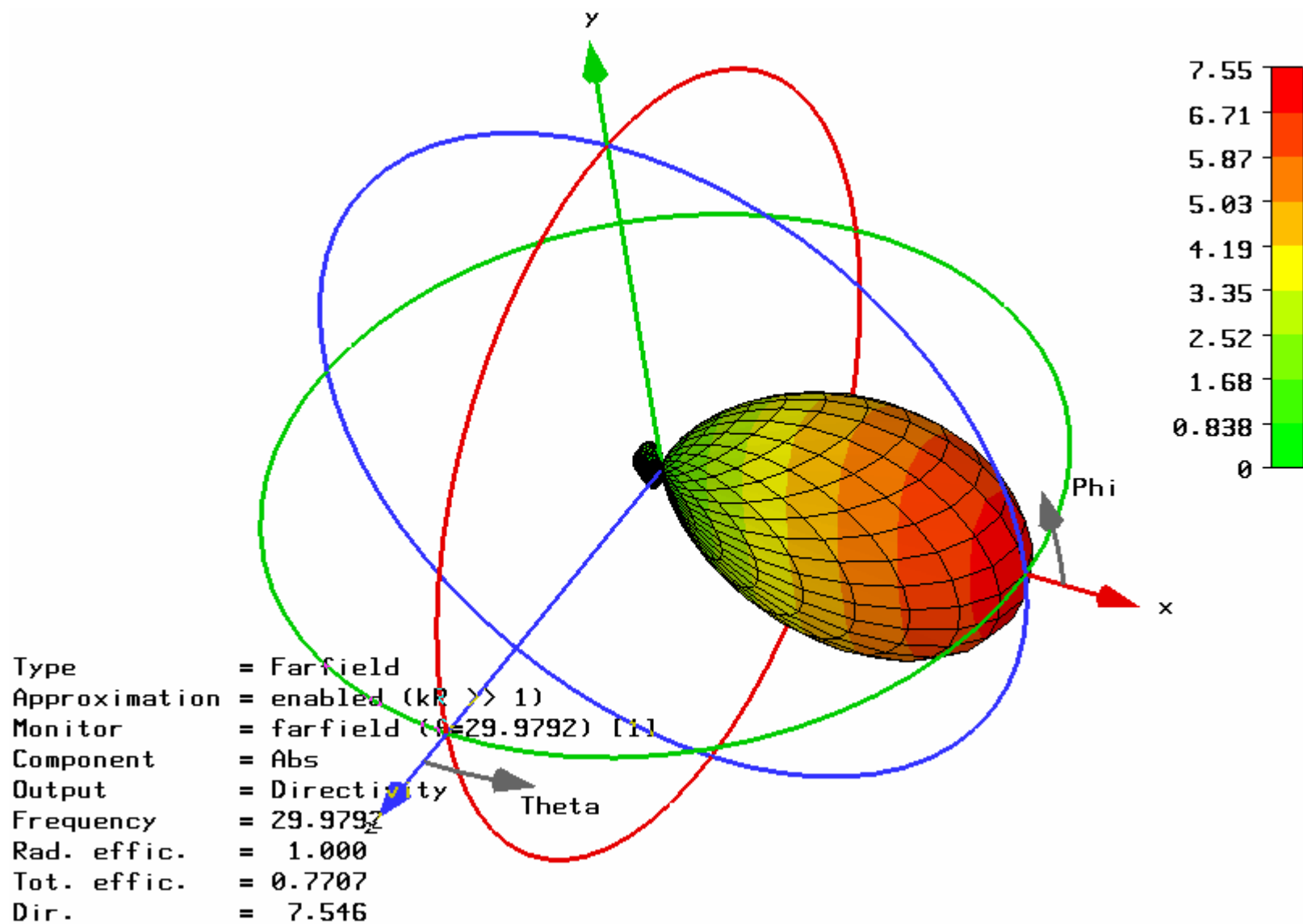
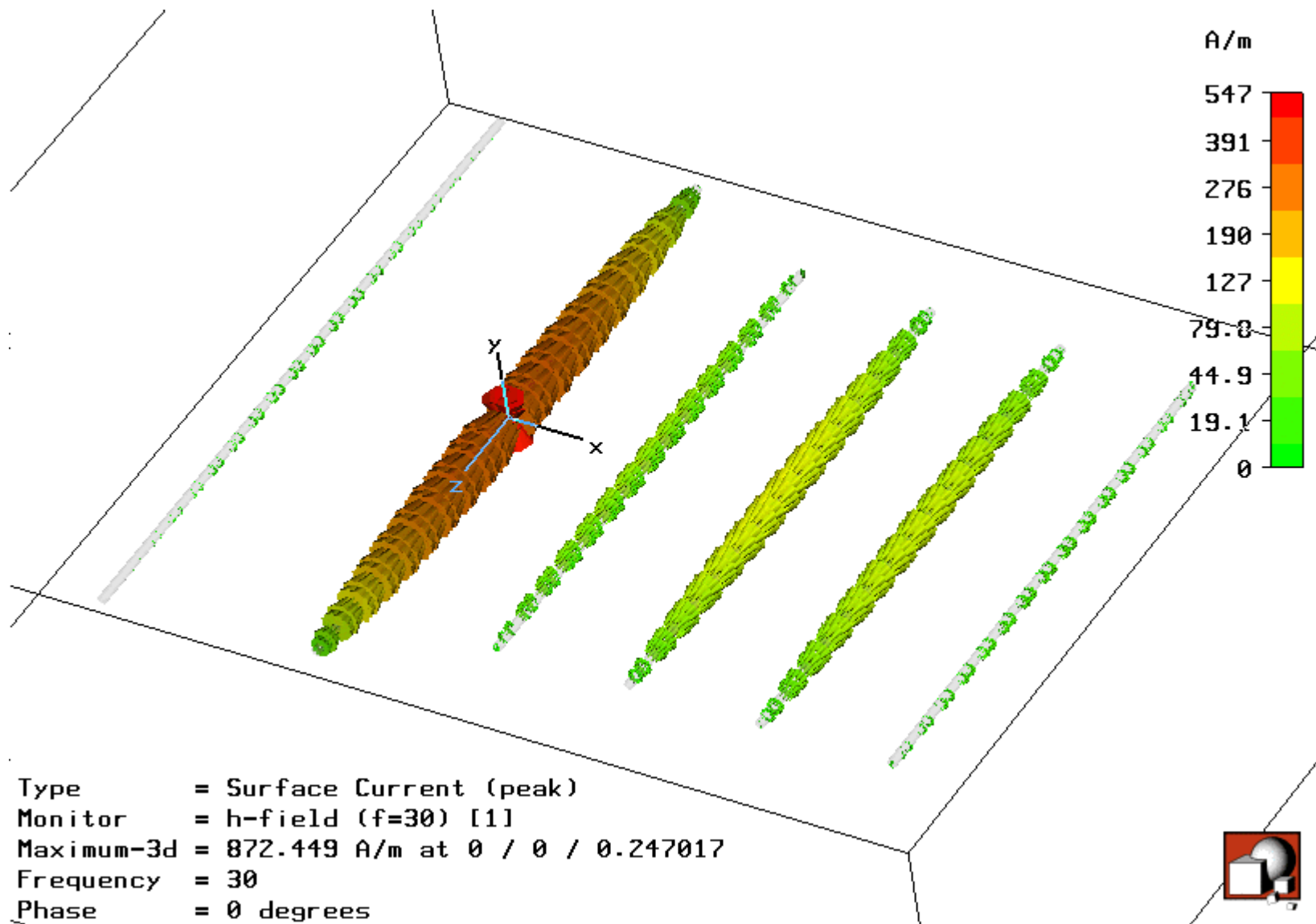


Figure 10.21 Directivity and front-to-back ratio, as a function of reflector spacing, of a 15-element Yagi-Uda array.

Kết quả mô phỏng một anten Yagi:







Anten Yagi với các chấn tử vòng

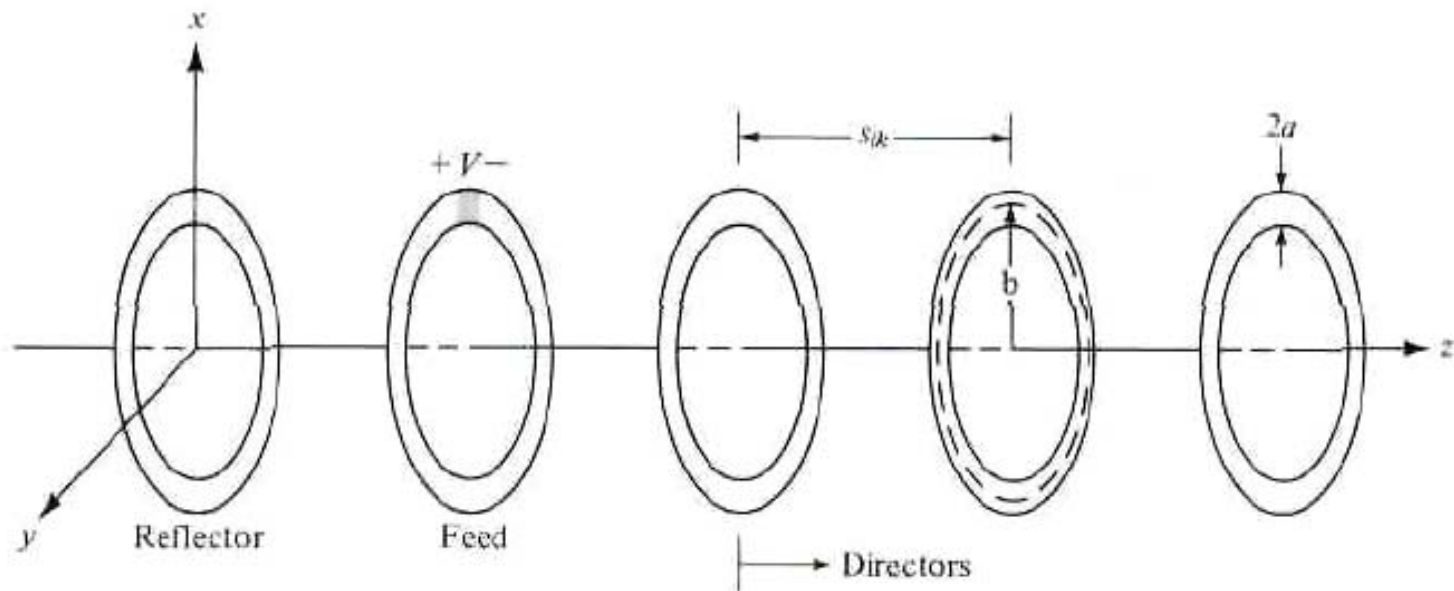


Figure 10.27 Yagi-Uda array of circular loops.

3. Anten Helic

Chiều dài anten: $L = NS$

Tổng chiều dài của dây xoắn:

$$L_n = N L_0 = N \sqrt{S^2 + C^2}$$

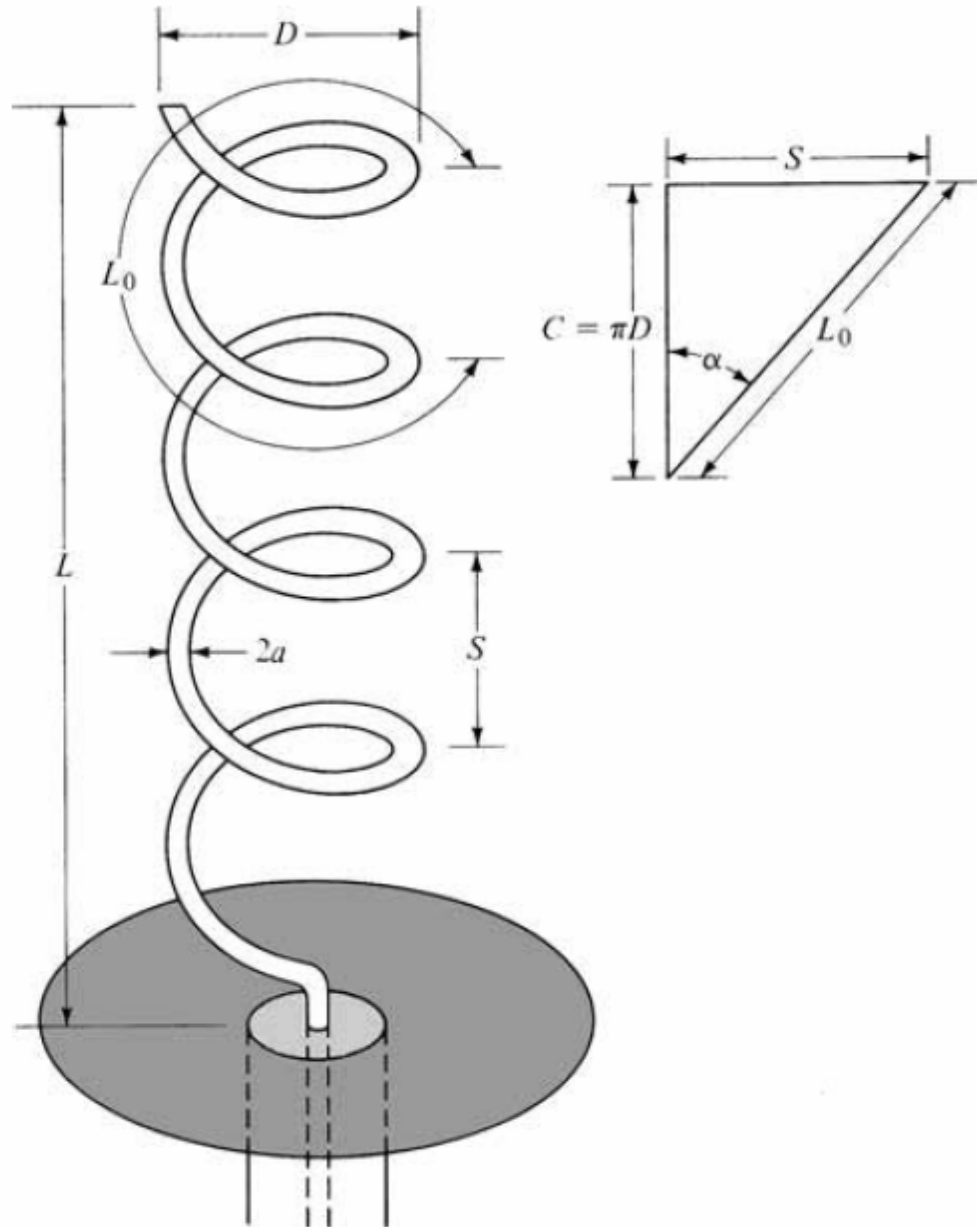
$C = \pi D$ - Chu vi của mỗi vòng

$$L_0 = \sqrt{S^2 + C^2}$$

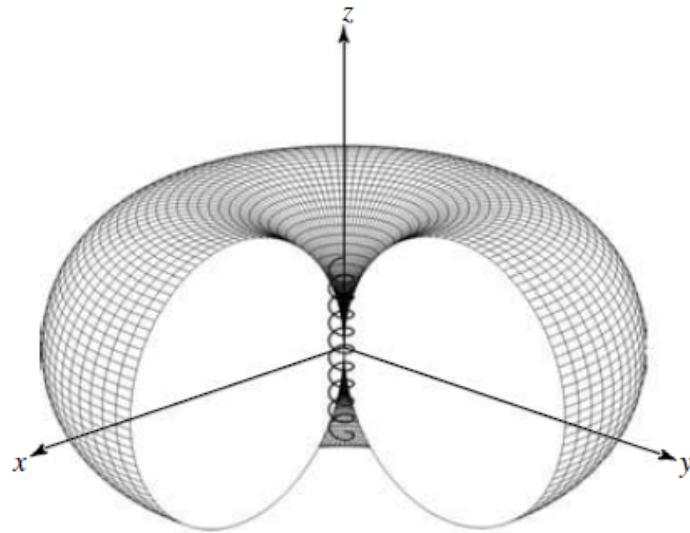
- Chiều dài dây giữa mỗi vòng

Pitch angle:

$$\alpha = \tan^{-1} \left(\frac{S}{\pi D} \right) = \tan^{-1} \left(\frac{S}{C} \right)$$



Normal mode:

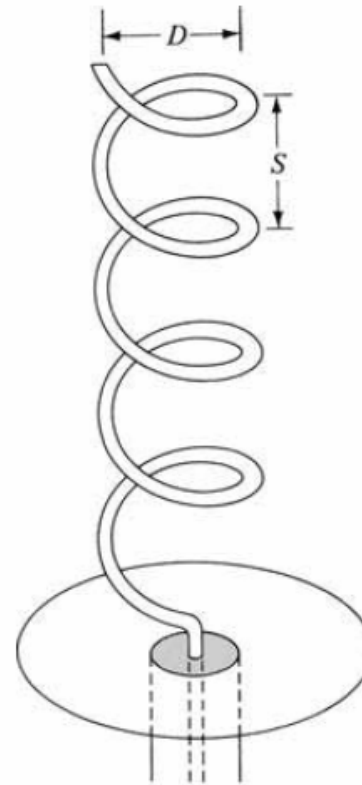


(a) normal mode

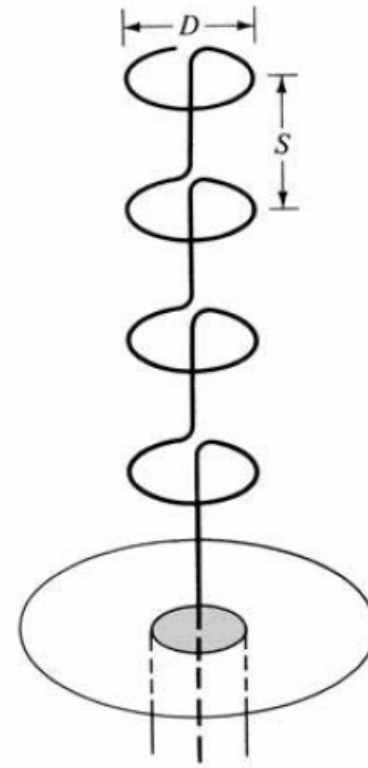
Trường bức xạ:

$$E_{\theta} = j\eta \frac{k I_0 S e^{-jkr}}{4\pi r} \sin \theta$$

$$E_{\phi} = \eta \frac{k^2 (D/2)^2 I_0 e^{-jkr}}{4r} \sin \theta$$



(a) Normal mode

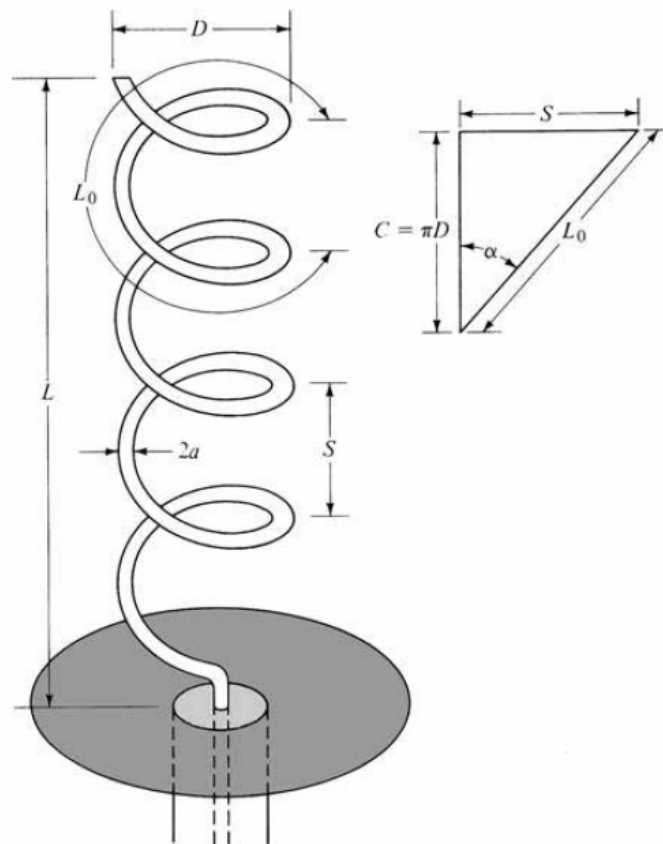


(b) Equivalent

Axial Ratio:

$$AR = \frac{|E_{\theta}|}{|E_{\phi}|} = \frac{4S}{\pi k D^2} = \frac{2\lambda S}{(\pi D)^2}$$

Axial mode:



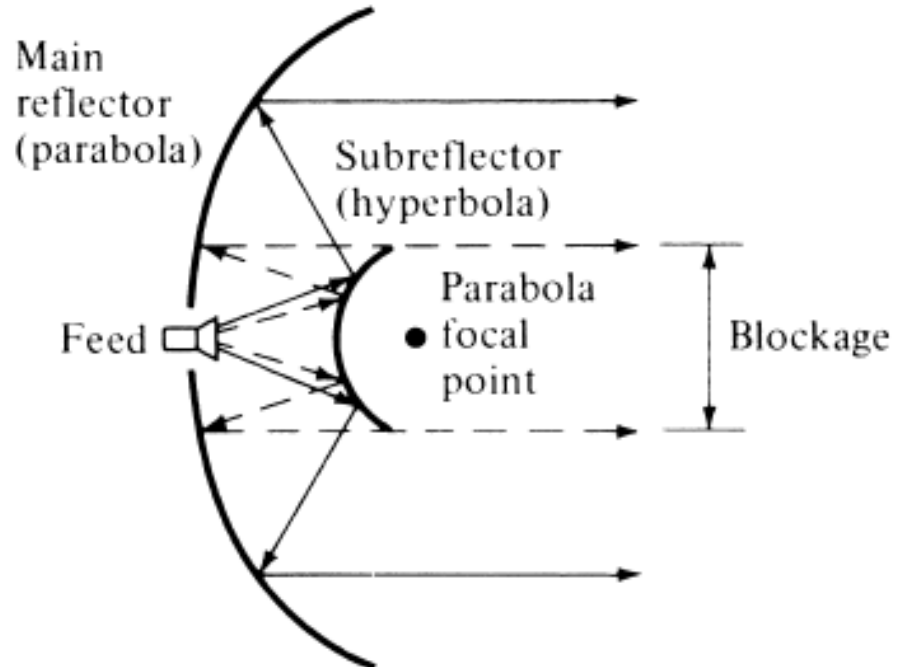
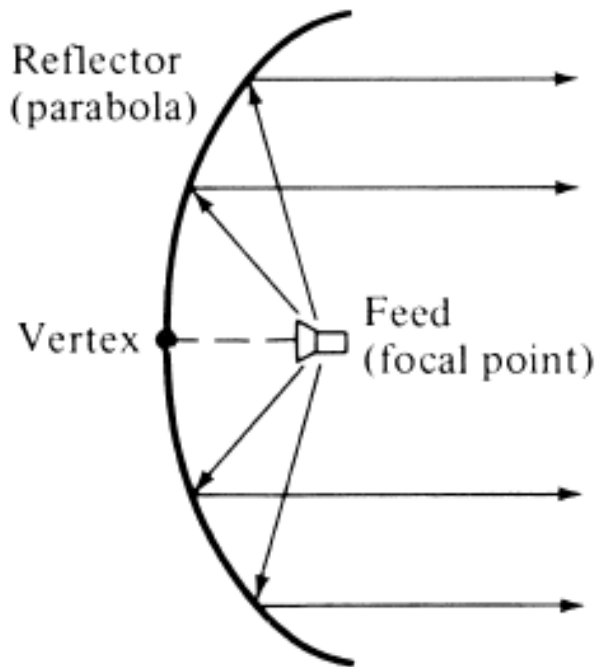
Để có phân cực tròn:

$$\frac{3}{4} < C/\lambda_0 < \frac{4}{3}$$

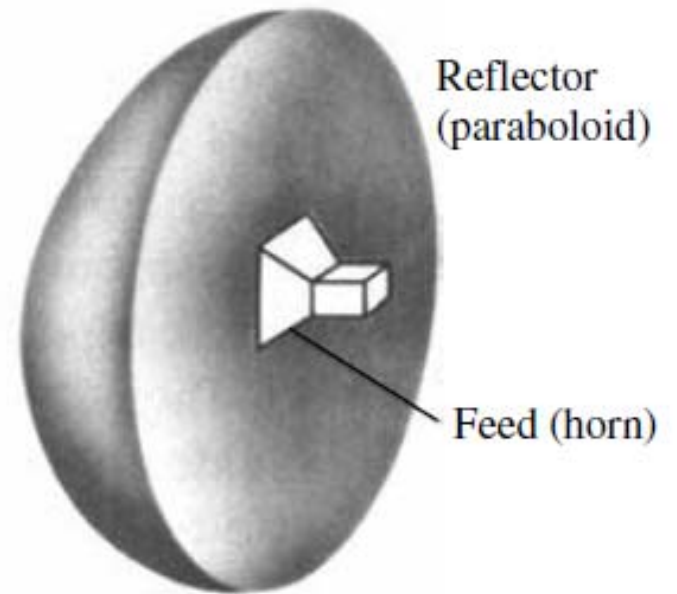
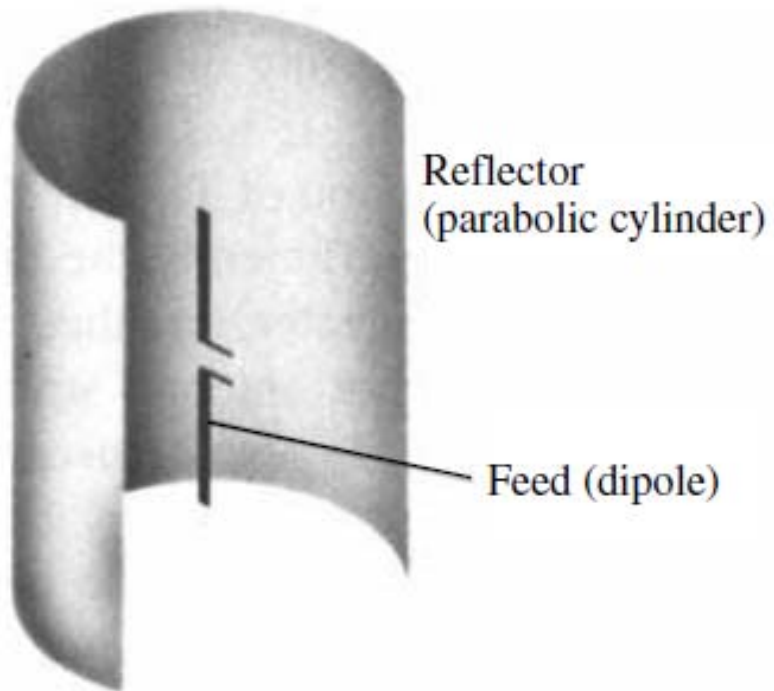
$$S \simeq \lambda_0/4$$

Thông thường $12^\circ \leq \alpha \leq 14^\circ$

4. Anten Parabol



Minh họa một số mặt parabol



Mặt phản xạ

$$OP + PQ = \text{constant} = 2f$$

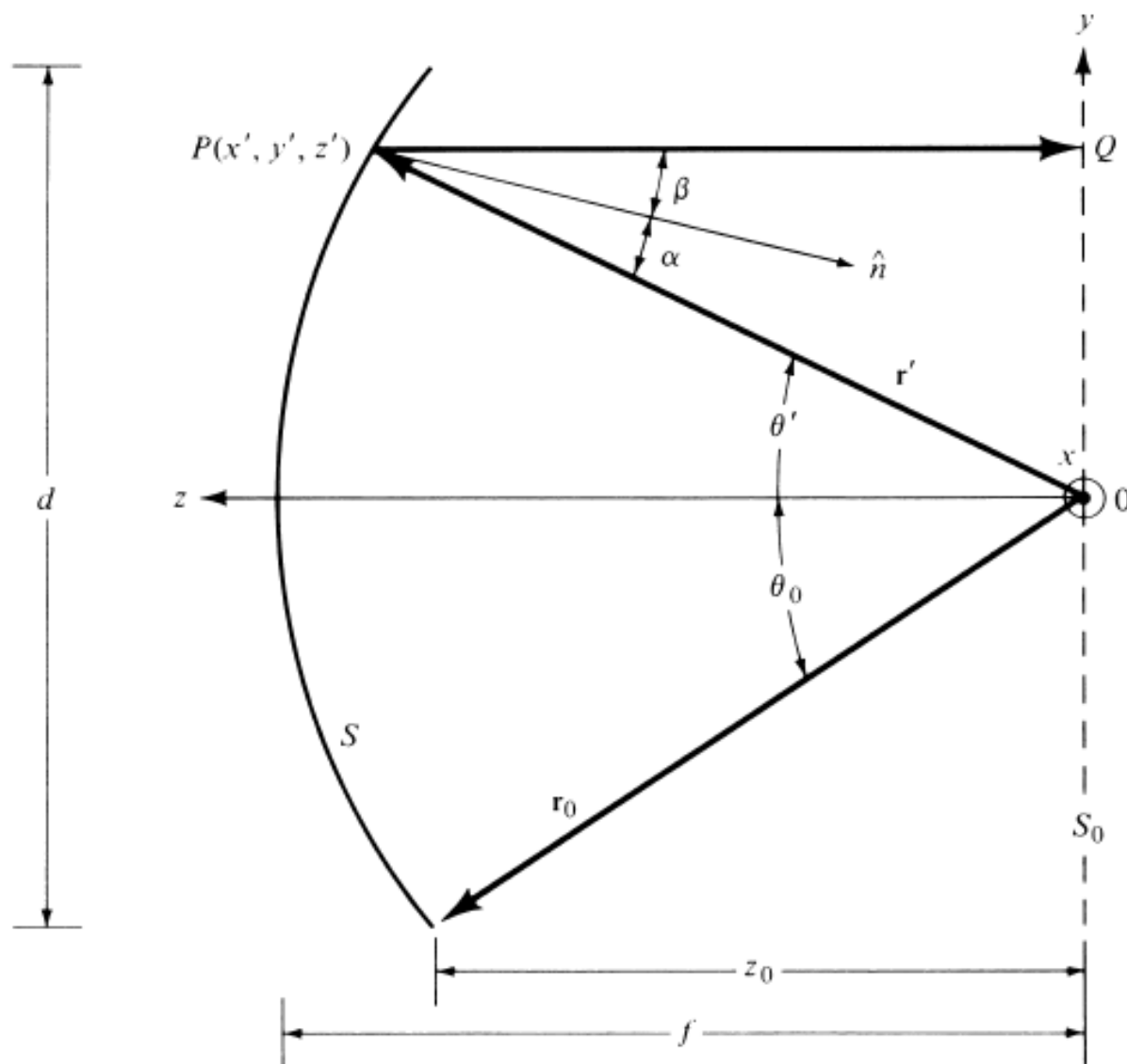
$$OP = r'$$

$$PQ = r' \cos \theta'$$

$$r'(1 + \cos \theta') = 2f$$

$$r' = \frac{2f}{1 + \cos \theta'} \quad \theta \leq \theta_0$$

$$\theta_0 = \tan^{-1} \left| \frac{\frac{d}{2}}{f - \frac{d^2}{16f}} \right|$$

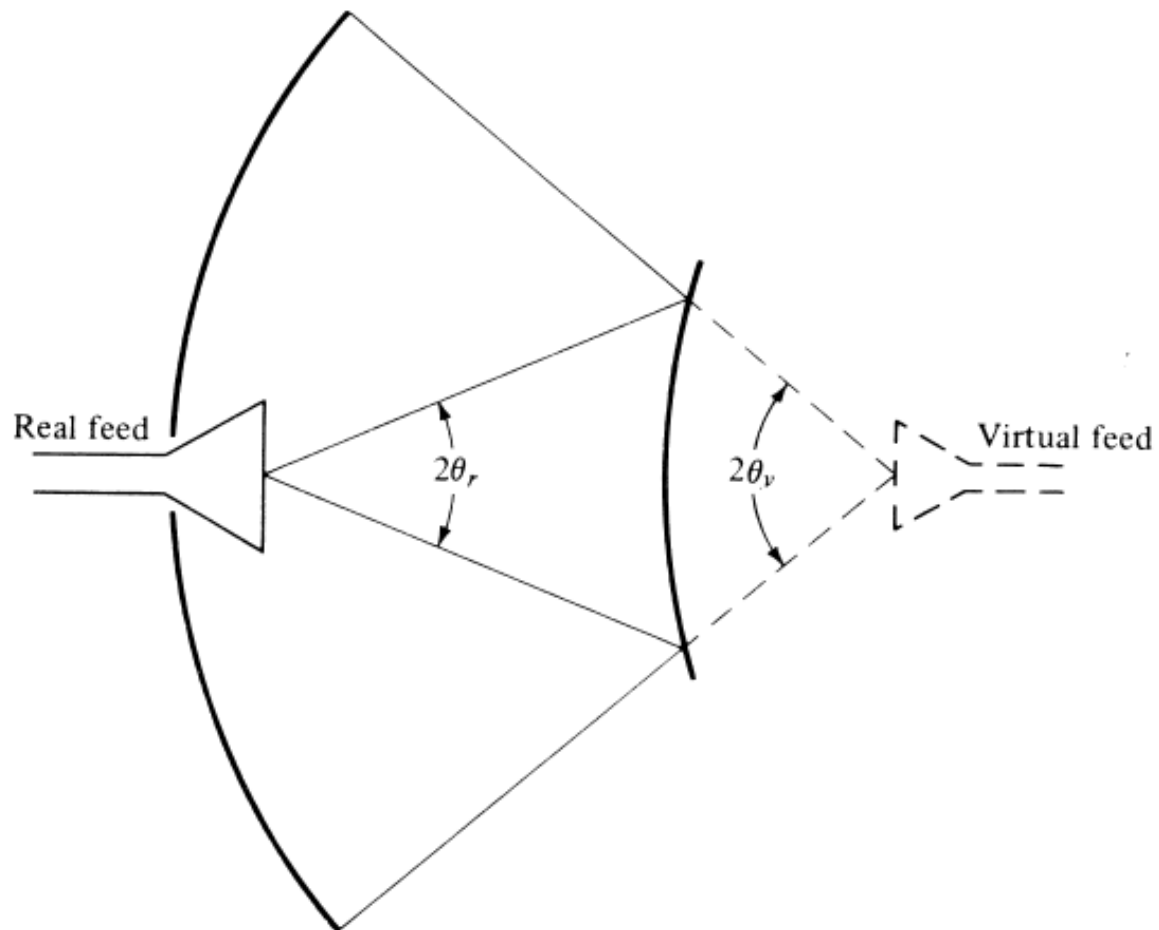


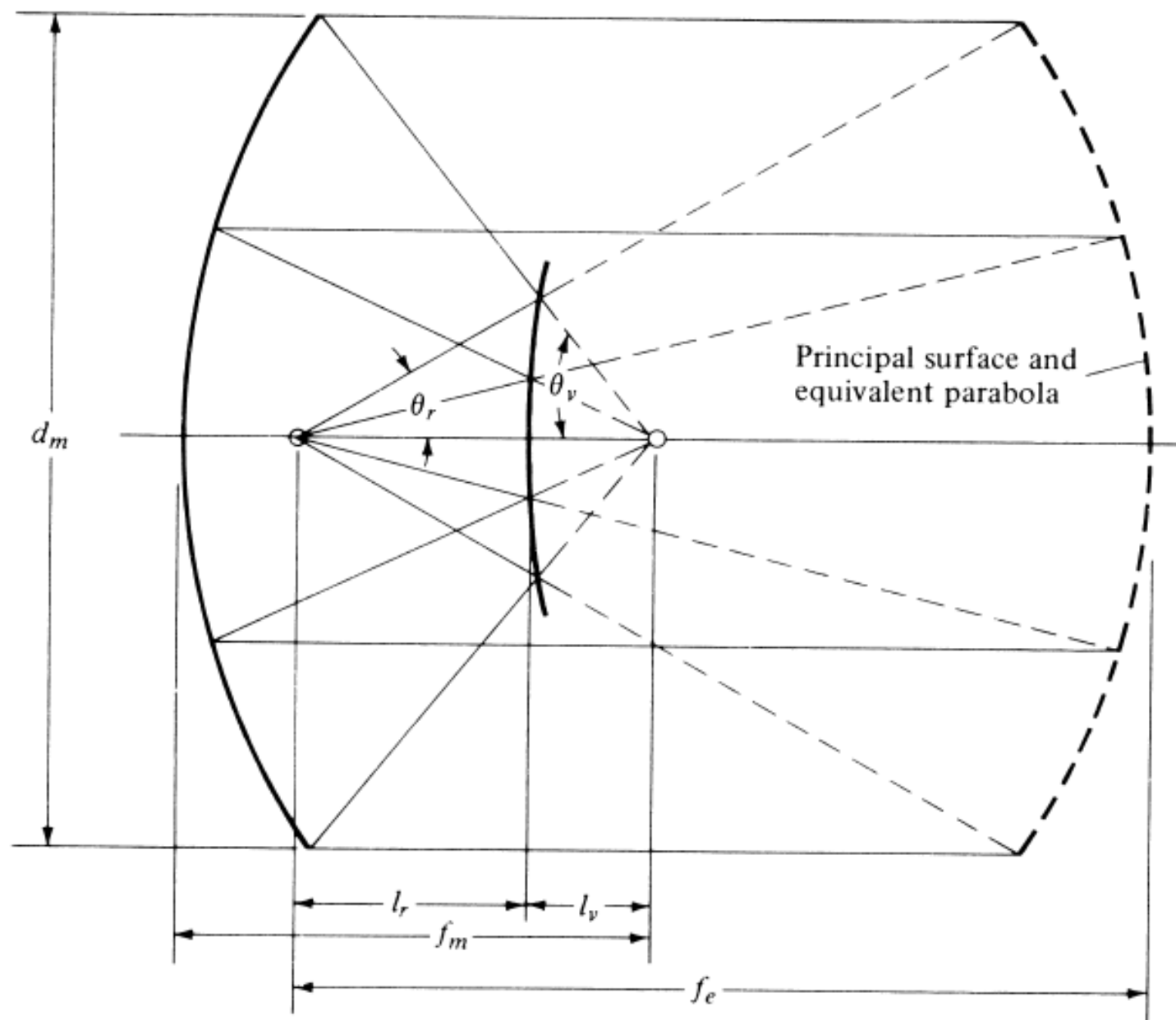
Hệ số định hướng :

$$D_0 = \left(\frac{\pi d}{\lambda} \right)^2 \left\{ \cot^2 \left(\frac{\theta_0}{2} \right) \left| \int_0^{\theta_0} \sqrt{G_f(\theta')} \tan \left(\frac{\theta'}{2} \right) d\theta' \right|^2 \right\}$$

Với G là hàm định hướng của bộ kích thích theo góc θ'

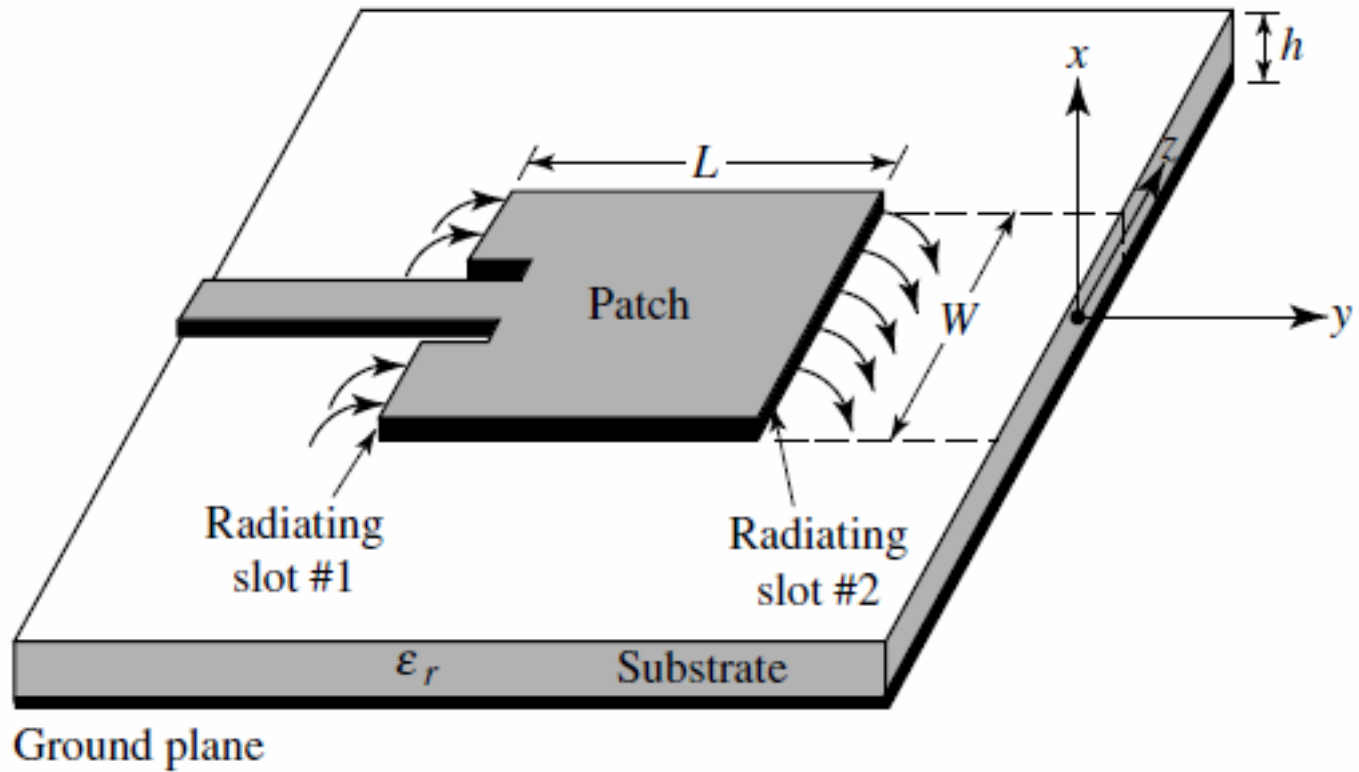
Anten parabol với mặt phản xạ phụ





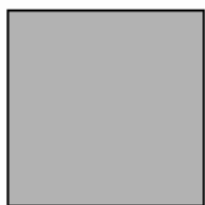
(b) Equivalent-parabola

5. Anten Vi dải



(a) Microstrip antenna

Hình dạng anten vi dải



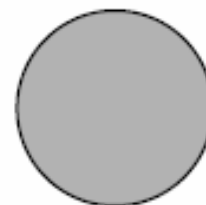
(a) Square



(b) Rectangular



(c) Dipole



(d) Circular



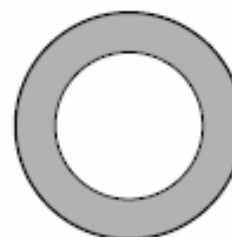
(e) Elliptical



(f) Triangular



(g) Disc sector

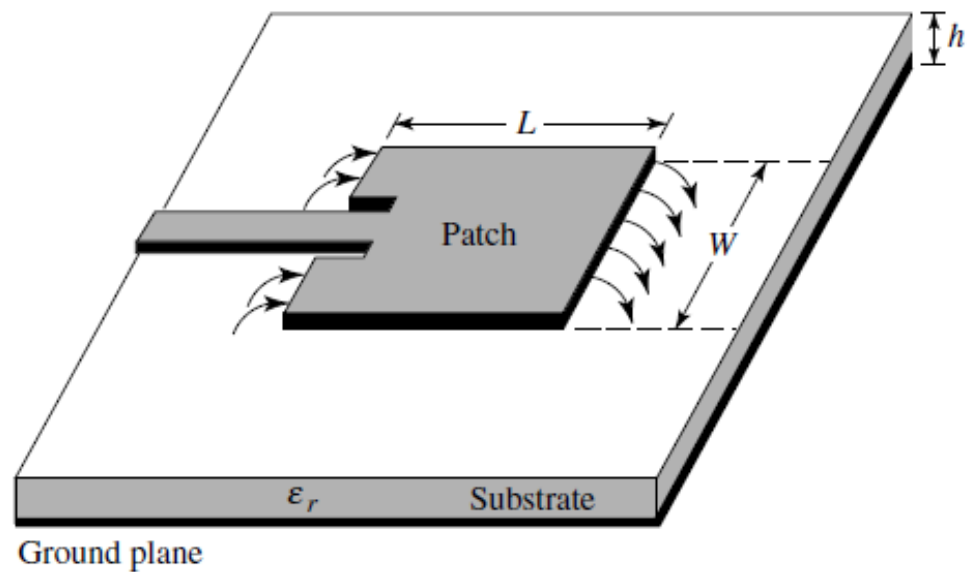


(h) Circular ring

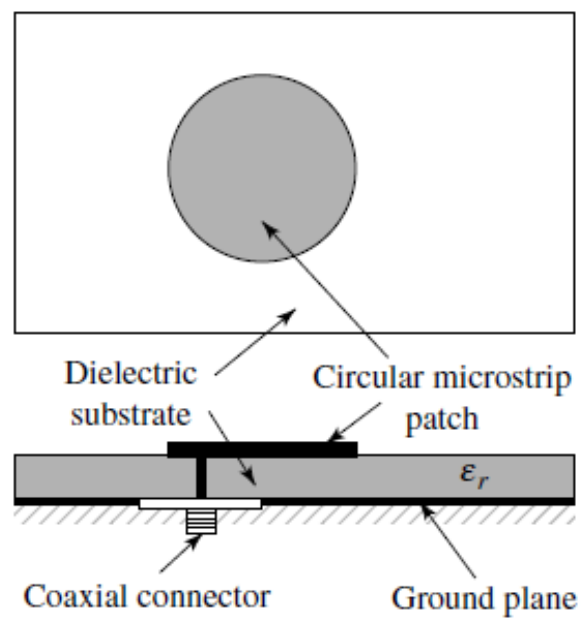


(i) Ring sector

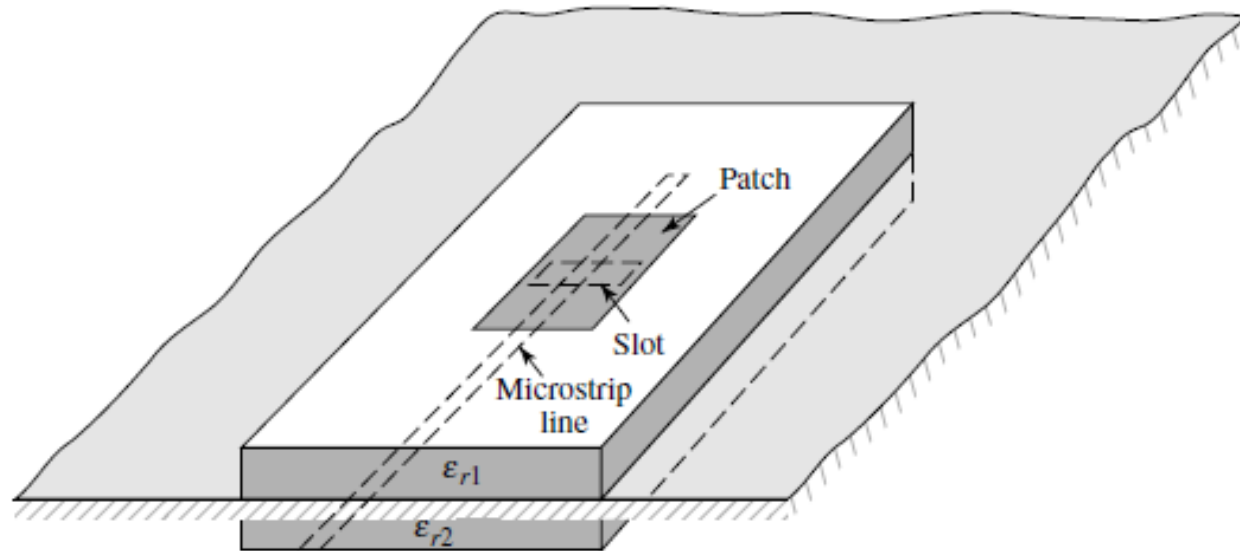
Kích thích anten vi dải



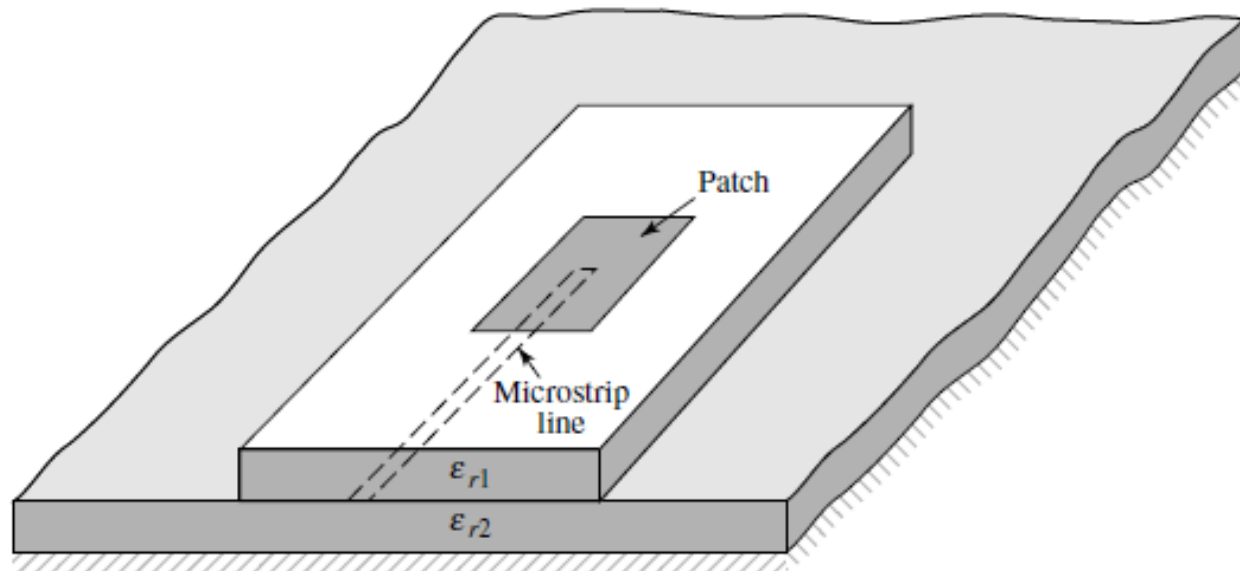
(a) Microstrip line feed



(b) Probe feed



(c) Aperture-coupled feed



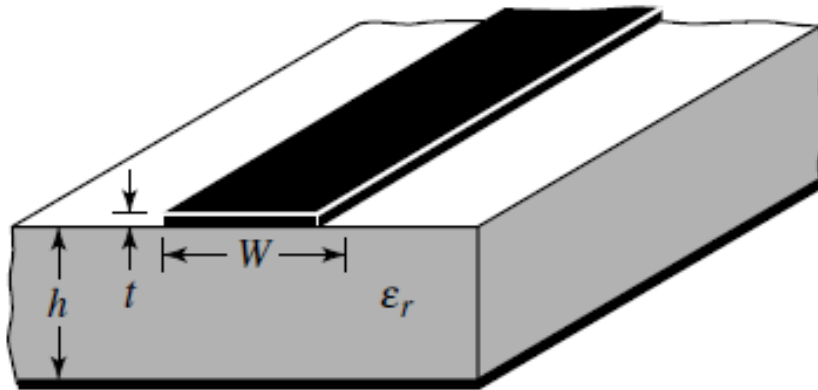
(d) Proximity-coupled feed

Phân tích anten vi dải:

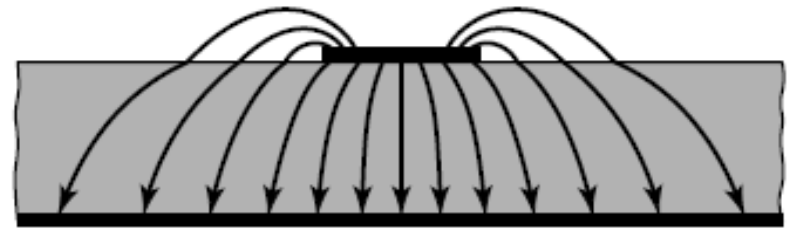
Mô hình đường truyền sóng

Mô hình hộp cộng hưởng

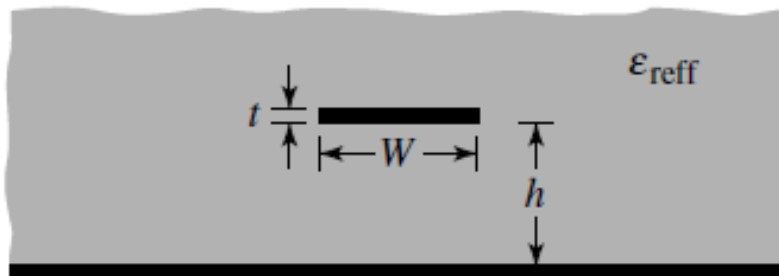
Mô hình đường truyền sóng



(a) Microstrip line

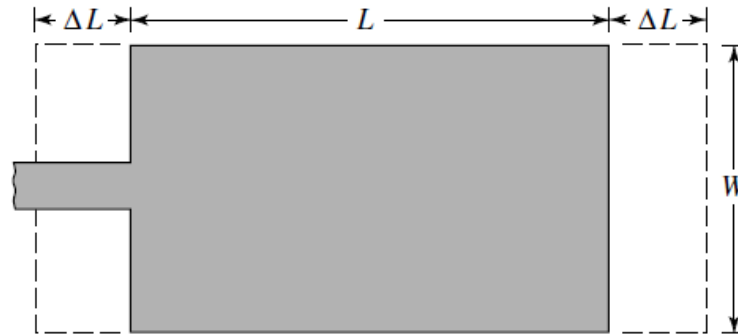


(b) Electric field lines

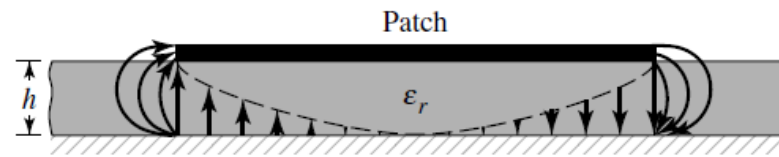


(c) Effective dielectric constant

$$\epsilon_{\text{reff}} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left[1 + 12 \frac{h}{W} \right]^{-1/2} \quad \text{for } W/h > 1$$



(a) Top view



(b) Side view

Figure 14.7 Physical and effective lengths of rectangular microstrip patch.

$$\frac{\Delta L}{h} = 0.412 \frac{(\epsilon_{\text{reff}} + 0.3) \left(\frac{W}{h} + 0.264 \right)}{(\epsilon_{\text{reff}} - 0.258) \left(\frac{W}{h} + 0.8 \right)} \quad L_{\text{eff}} = L + 2\Delta L$$

$$W = \frac{1}{2f_r \sqrt{\mu_0 \epsilon_0}} \sqrt{\frac{2}{\epsilon_r + 1}} = \frac{v_0}{2f_r} \sqrt{\frac{2}{\epsilon_r + 1}}$$

Mô hình hộp cộng hưởng

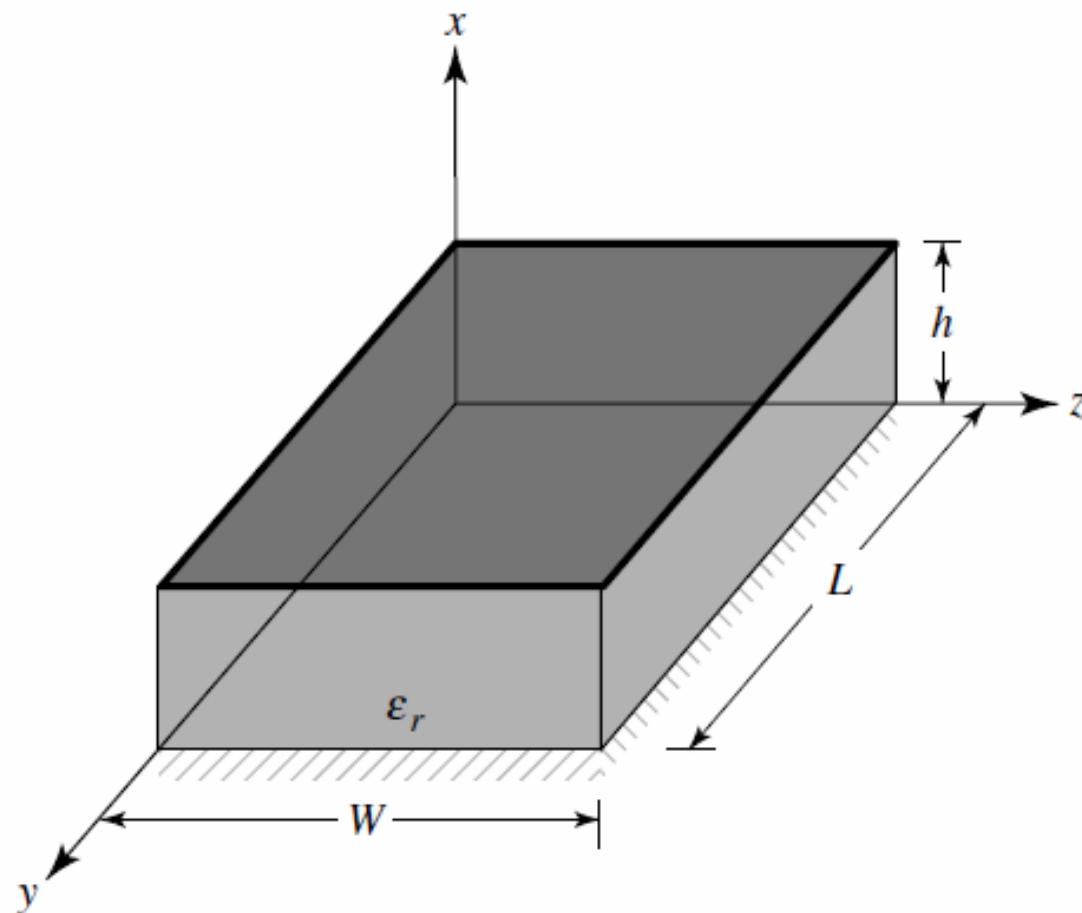
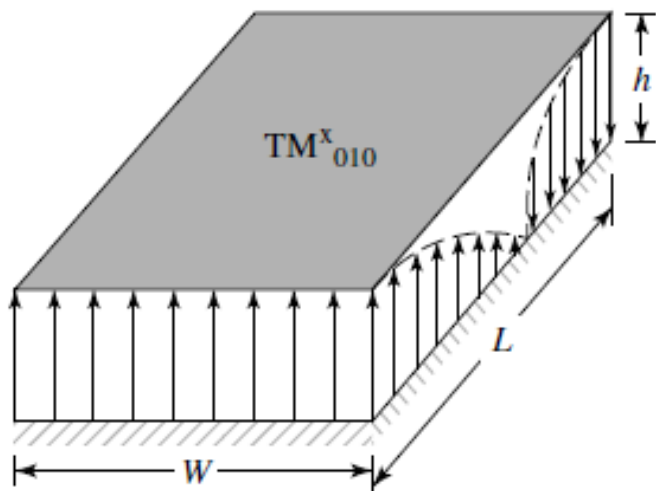
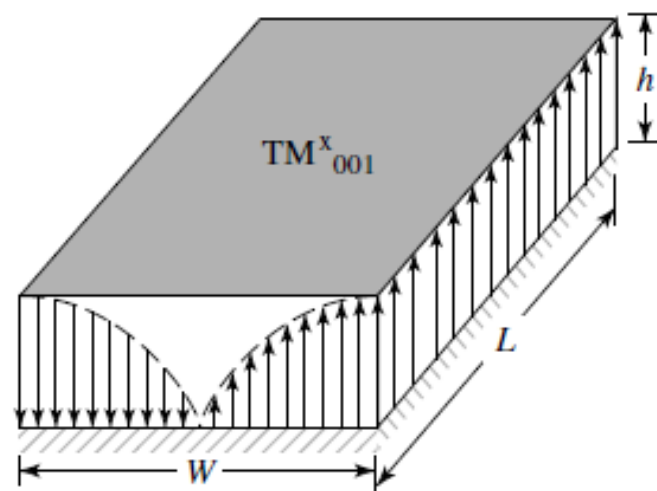


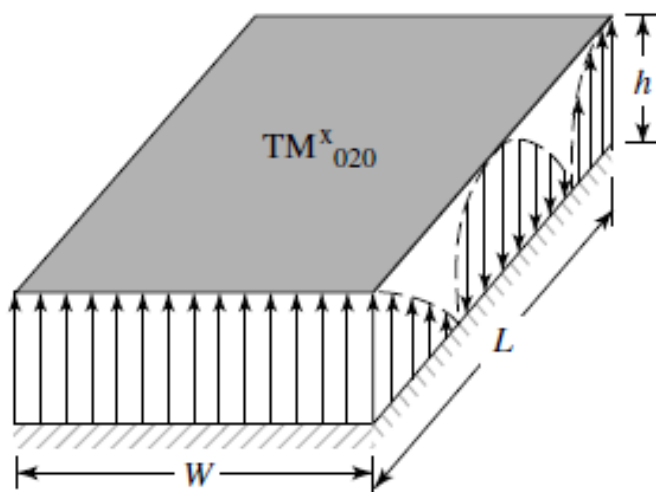
Figure 14.13 Rectangular microstrip patch geometry.



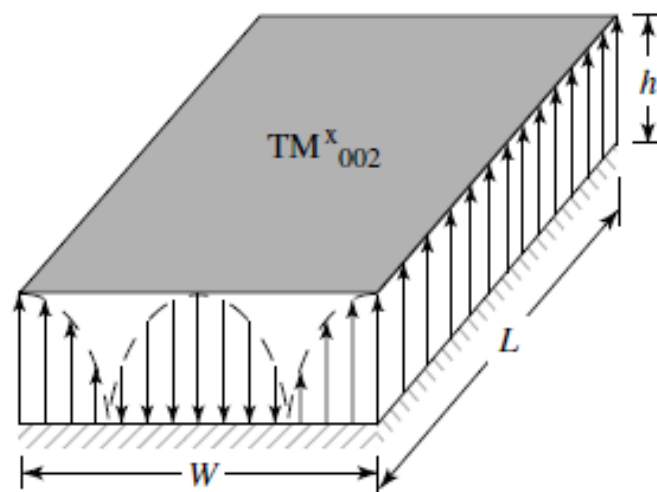
(a) TM^x_{010}



(b) TM^x_{001}



(c) TM^x_{020}



(d) TM^x_{002}

Figure 14.14 Field configurations (modes) for rectangular microstrip patch.

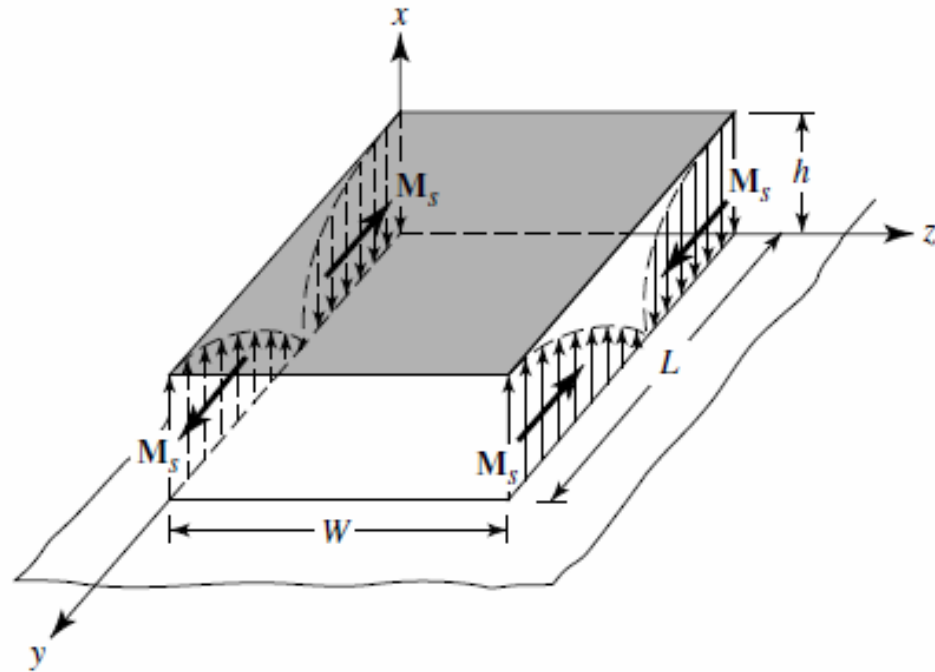
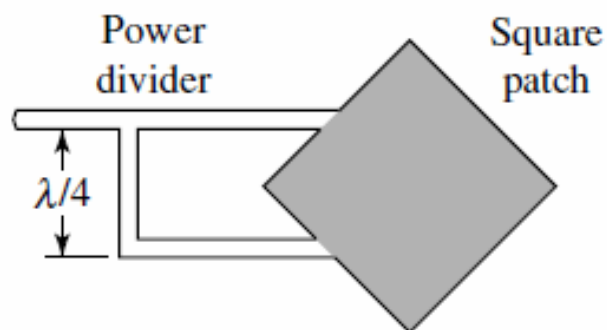
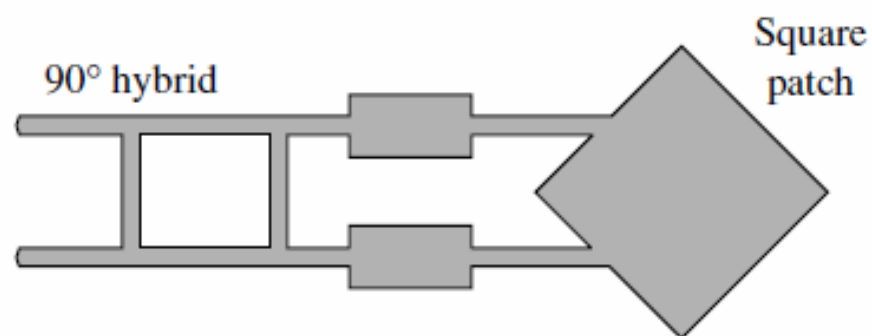


Figure 14.18 Current densities on nonradiating slots of rectangular microstrip patch.

Anten vi dải với phân cực tròn



(a) Square patch driven at adjacent sides through a power divider



(b) Square patch driven at adjacent sides through a 90° hybrid